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Minimal Surfaces, Measure-Based Metric and Image Segmentation

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Abstract

One of the primary goals of low level vision is image segmentation: given data g, defined as a function on the "pixel space" B, the objective is to deduce an image u which is composed of sub-domains, wherein the image is basically homogeneous, separated by a sharp discontinuities (edges).

It has been shown that a large number of algorithms for image segmentation are closely related to the Mumford-Shah functional minimization [42]. This functional involves a tradeoff between the image structure, which is a two-dimensional surface, and the contours that surround objects or distinct regions in the image, which are one-dimensional parametric curves. This functional was first suggested and analyzed in its one-dimensional case by Mumford and Shah for gray level images [41]. The above functional was later extensively studied, (see e.g. [40] for an overview).

In particular, the Γ-convergence framework [46] was invented to overcome the problem of dealing with objects with different dimentionalities in the same functional. The idea is to approximate the functional by a different, parameter dependent functional, that is expected to be more regular. The approximating functional approaches the original one in the limit, while the parameter goes to zero. According to this approach, minimizers of approximating functional approximate the minimizer of original one, while enjoying greater regularity.

In this study we propose an alternative functional to Mumford-Shah's one. The proposed functional is independent of parameterization; it is a geometric functional which is given in terms of the geometry of surfaces representing the data and image in the feature space. The Γ-convergence technique is merged with the minimal surfaces theory in order to yield a global generalization of the Mumford-Shah segmentation functional.

Chapter 1

Introduction

Natural and/or man-made visual environment is usually perceived as being endowed with a certain structure, often associated with objects. The question is: how is the stream of unstructured visual data, impinging on the retina, perceived as a structured image?

This question has been addressed in a scientific framework by the Gestalt school of psychologists in the twenties, and then, in the eighties, by psychophysicists who assessed how much may be perceived by human being in the very short interval of time. At the end of the sixties the same problem was addressed in a very practical framework by engineers who wished to extract automatically useful information from digital images. The goal of this effort is to build perception-driven robots and to better understand the human vision.

David Marr [36] inspired by his model the following subdivision of vision: low-, intermediate and high-level, each concerned with its own level of image description. Low level vision seeks the description in terms of edges, ridges, fragments of lines. Intermediate level is concerned with objects, their geometry and background. High-level vision attempts to recognize objects and to obtain a full scene description. *Image Segmentation* is a problem associated with all the levels of vision.

To proceed, we assume that a three-dimensional physical environment is observed by an eye or an electro-optical imaging system from some point P. If one has a lens at P focusing the stream of photons on an electro-optical sensor, a planar image domain B with coordinates (x, y) may be introduced. Let $g(x, y)$ be the intensity of the light signal impinging on B at a point (x, y). The function $g(x, y)$ defined on the plane domain B will be called an *image*.

Light reflected from the surfaces S_i of various solid objects O_i, visible from P, will be projected onto various open subsets B_i of the domain B. When the object O_1 is partially in front of another object O_2 as seen from P, and part of object O_2 appears as the background of O_1, then the open sets B_1 and B_2 will have a common boundary, defining an *edge* of object O_1. The image $g(x, y)$ is expected to be discontinuous along such edges. Other discontinuities in g are caused by discontinuities in the

surface orientation of visible objects, discontinuities in objects' albedo (i.e., surface markings) and discontinuities in illumination (e.g. shadows). Including all these effects, the image $g(x, y)$ is expected to be piecewise smooth, i.e. it is well modelled by a set of smooth functions u_i defined on a set of disjoint regions B_i covering B. We shall adopt this relatively simplistic piecewise smooth model, although it does not account for textured objects, partially transparent and reflecting objects, and shadows, since it is applicable to a broad range of real-life problems. Further, this model facilitates the formulation of the *segmentation problem*, being defined in the informal way by Haralick and Shapiro [24], or by Ullman [54] with the focus on the goal of recognition.

The definition of the segmentation also depends on whether one approaches the problem at the level of the image as a whole, where it is then natural to consider the partitioning of the image into smaller structures or, alternatively, considers the image as a collection of edge fragments, in which case it becomes more natural to consider the problem of grouping the elements into larger structures. In any case, the following questions arise: (*i*) What exactly is the goal of the segmentation process? and (*ii*) Is segmentation feasible? These questions are important for the understanding of the above process and, in fact, without a clear conception of the task and its requirements no satisfactory progress in this area can be made.

Indeed, the obvious fact that segmentation criteria can vary significantly indicates that a general theory of segmentation should not focus primarily on the details of those criteria. Thus, the last assertion leads to following questions:

(**a**) Do segmentation criteria have a generic form, which could be therefore studied as a whole?

(**b**) Is there a computational theory to automatically find a compromise between those competing criteria?

Morel and Solimini [40] showed that any heuristic segmentation method may be translated into a variational one. Variational formulation consists of summarizing all criteria concerning a set of edges K in a single functional $F(K)$ with real value. In other words, to any set of edges or "segmentation" K is associated a value $F(K)$ which states how "good" the segmentation is. A classical convention is to define F in such a way that $F(K)$ is better the lower it is. This fact lends itself to an obvious comparison principle: given two segmentations, it must determine which is better. In other words, a variational method implicitly orders all segmentations. Any order can be quantified by $F(K)$ such that $F(K) > F(K')$ if K' is better than K. It is not a restrictive strong assumption that $F(K)$ is a real number.

The present study merges the Γ-convergence technique (see [46]) and the minimal surface theory (see [18]) to yield a global generalization for the Mumford-Shah segmentation functional. This approach results in a segmentation algorithm, suitable for any kind of images (grey level, color, texture, etc.).

This work is organized as follows:

In Chapter 2 we review the existing methods for image segmentation. In particular, we summarize the first attempts in image segmentation and edge detection - non-variational methods, the Mumford-Shah segmentation functional ([42]), which served as a basis for development of the functional proposed in our study, and the various dynamic models for edge detection and image segmentation, which introduce the different contour evolution algorithms. In addition, we review the unified (Beltrami) framework for low-level vision, including image segmentation (see [51]), and a couple of works including the Γ-convergence technique (see [46]) because of its importance in the derivation of our image segmentation in Chapter 3.

In Chapter 3 we propose an alternative to the Mumford-Shah functional. The concept behind this alternative is a geometric functional which is independent of parameterization, and can be implemented in terms of the geometry of surfaces representing the data and image in the feature space. The Γ-convergence technique is merged with the theory of minimal surfaces in order to yield a global generalization for the Mumford-Shah segmentation functional. Starting with its original version given in [42], we bring the full construction of our functional. The main contribution of this chapter is the definition of absolutely new type of measure based metric, $\mathcal{D}(g, u)$, representing the distance between the two surfaces , and its subsequent relaxation (using Γ-convergence technique) into a form which is convenient for numerical implementation.

In Chapter 4 we treat the minimization problem for relaxed functional with parametric representation. In particular, we provide the above functional's first variation and prove the $\mathcal{C}^\infty$ interior regularity of its minimizer. Eventually we conclude some properties concerning the interior regularity of minimizer of the original functional.

In Chapter 5 we present a detailed solution of the minimization problem for the relaxed functional with non-parametric representation. In particular, we provide the above functional's first variation and discretize the one- and two-dimensional cases in order to obtain the numerical solution. We illustrate the results with suitable numerical experiments: 2 examples of one-dimensional case in which we compute the segmentation functions for Gaussian interlaced with piecewise constant function, and for Gaussian perturbed by a highly oscillating function, both with different parameters. Finally, we illustrate the results with real grey-level and medical ultrasound images, and discuss the effects of the parameters.

Discussion and conclusions concerning the results of this work are given in Chapter 6.

The proofs of regularity theorems from Chapter 4 are given in Appendices A and B.

Chapter 2

Review of background

2.1 Introduction

We first outline a brief history of image segmentation and edge detection algorithms. Segmentation is one of the important tasks in image analysis and a great deal of effort has been devoted to its solution. The early algorithms were relatively naive and unsophisticated: some of them (like Prewitt and Sobel edge detectors) tried to locate points defined by local maxima of the image gradient over a neighborhood $\mathcal{N}(x, y)$ of the pixel (x, y). Others were based on grey level threshold selections. Such methods that were developed from the early sixties till the end of seventies were widely addressed by Rosenfeld and Kak [47]. Another techniques like spatial clustering, region growing schemes, split-and-merge schemes were reviewed by Haralick and Shapiro [24].

More advanced algorithm was proposed by Marr and Hildreth [37]. They considered as an edge detector the zero crossings of the Laplacian of the image. Since it was also considered wise to smooth noise by a convolution with Gaussian kernel, the edges of Marr and Hildreth were defined as the zero crossing curves of a Laplacian of Gaussian (LoG) operator $\nabla^2 G$ applied to the image. Haralick in [23] considered as an edge detector the zero crossings of the second order derivative along the image gradient direction. Canny in [8] attempted to motivate Haralick edge detector by important observation that using only the gradient direction component of the Laplacian can decrease the influence of additive random noise due to the elimination of the second order derivative along the level sets that does not contribute to its localization. Eventually, a two-dimensional variational explanation for the Marr-Hildreth and the Haralick-Canny edge detectors was recently presented in [29] and [30]. Zero crossings of image Laplacian were shown to be optimal edge integration curves solving a minimization problem, whose normals best align with the image gradient field. Its only difference is the additional minimization term, whose role is to group together uniform areas, avoiding noisy regions, and thereby having a chance to yield more robust edge curves for better segmentation.

2.2 Mumford-Shah functional

The Mumford-Shah variational model [42] has become a general framework for image segmentation. It defines the segmentation problem as a joint smoothing/edge detection problem as follows: given an image g, it seeks simultaneously a set K of discontinuities, the "edges" of g, decomposing the planar domain B ("pixel space") to disjoint connected open subsets B_i, $i = 1 \dots n$, so that $B = B_i \cup \dots \cup B_n \cup K$, and a function u differentiable on $\cup B_i$, which is allowed to be discontinuous across K. Then, the "best" segmentation of a given image is obtained by minimizing the functional

$$F(u, K) = \alpha \int_B (u - g)^2 dxdy + \int_{B \setminus K} |\nabla u|^2 dxdy + \beta |K|, \tag{2.1}$$

where $|K|$ stands for the total length of the arcs making up K and α, $\beta > 0$ are two weight parameters. The smaller F is, the better (u, K) segments g. The first term accounts for how good is the approximation of g by u. It is referred to as *fidelity* term. The second, *smoothing* term, makes sure that u - and hence g - does not vary much on each B_i. The third term is a penalty on the total length of the discontinuities.

It isn't obvious that the problem of minimizing F is well posed. Mumford and Shah conjectured that for all continuous functions g, F has a minimum in the finite set of all pairs (u, K), u differentiable on each B_i, K a finite set of $\mathcal{C}^1$ curves. So far, this conjecture has not been fully proved and only partial but meaningful enough results are at hand. The problem has turned out difficult for the presently-employed mathematical technique, because of the subtle interaction of the two-dimensional term (in u) and the one-dimensional term $|K|$.

As was noted above, the fundamental problem is to find u and K which minimize the value of F. Fixing K, standard calculus of variations shows that F is a positive definite quadratic functional in u with a unique minimum, that solves the elliptic boundary value problem on each B_i:

$$\nabla^2 u = \alpha(u - g), \tag{2.2}$$

$$\left.\frac{\partial u}{\partial \vec{n}}\right|_{\partial B_i} \equiv 0. \tag{2.3}$$

Here ∂B_i is the boundary of B_i, and $\partial / \partial n$ is a unit normal vector to ∂B_i. Assume B_i is partially smooth, then the above boundary value problem, (2.2)-(2.3), determines a unique solution $u \in \mathcal{C}^1$ with its second derivatives in L_p for all $p < \infty$. The operator $\alpha - \nabla^2$ ($\alpha > 0$) is a positive-definite selfadjoint operator and there is a Green's function $G(x, y; v, w)$ which is $\mathcal{C}^\infty$ except on the diagonal $(x, y) = (v, w)$ such that, for each component B_i of $B \setminus K$,

$$u(x, y) = \alpha \iint_{(v,w) \in B_i} G(x, y; v, w) g(v, w) dvdw$$

satisfies (2.2) and (2.3).

Writing u_K for this solution, F reduces to a function of K alone: $F(u_K, K)$. Next, a solution of infinitesimal variation problem of K yields a 2nd order ordinary differential equation for every $\mathcal{C}^{1,1}$ curve $\gamma_i \in K$:

$$e(u_K^+) - e(u_K^-) + \beta \cdot \text{curve}(\gamma_i) = 0 \quad \text{on } \gamma_i,$$

where

$$e(u_K; x, y) = \alpha(u_K(x, y) - g(x, y))^2 + \|\nabla u_K(x, y)\|^2,$$

$u_K^{\pm}$ denote boundary values of u_K along γ_i and curve(γ_i) is the curvature of γ_i. The terms $e(u_K^+)$ and $e(u_K^-)$ can be simply interpreted as the *energy density* corresponding to functional F above and below the curve γ_i.

Morel and Solimini [40] showed that many other segmentation algorithms are particular cases of Mumford-Shah functional or closely related to it. Unfortunately, minimizing the Mumford-Shah functional is a very difficult task.

2.3 Dynamic models for segmentation problem

In many cases there exists an algorithm providing an initial guess of the object boundary. There is a need in refinement procedure to apply it on initial (coarse) guess in order to adjust it to more appropriate location. For this purpose dynamic models ("snakes", geodesic active contours), using local information along the deforming curve, have been developed.

Originally, the "snakes" model, introduced by Kass, Witkin and Terzopoulos [27], was determined by an energy functional, including *external constraint forces* term, *internal forces* term and *image forces* term. The snake method provided an accurate localization of edges near a given initialization of the curve. However, there was no definite answer whether there exists stable and convergent algorithm for its implementation. Moreover, the snake model did not permit the simultaneous treatment of several contours. To conclude, the snake model is not an intrinsic model: for different parameterizations of the same curve one could obtain different final results. Caselles et. al. [9] introduced a geometric flow segmentation model, overcoming successfully problems characteristic of snake method. However, the above flow was not a minimization of meaningful functional. Caselles et. al. suggested a unification of both the parametric and the level set geometric model, known as the geodesic active contours [10]. The model functional was actually a new length formulation obtained by weighting the Euclidean element of length, which contains information regarding the boundary of object. The above flow deforms the initial curve (contour) towards the path of minimal weighted length. Eventually, in order to accelerate the motion of the geodesic active contours, Goldenberg et. al. [21] proposed to add an additional force that comes from area minimization term and motivated by the balloon force [17]. Thus, the contour may be directed to propagate inwards by minimization

of the interior.

An attempt to merge the active contours technique and Mumford-Shah functional theory was made by Chan and Vese [15], who proposed a minimal variance criterion (edgeless active contours). The optimal curve would best separate the interior and the exterior with respect to their relative average values. This model can detect contours without gradient and interior contours, automatically. It was therefore natural to extend it to vector-valued images. Chan, Sandberg and Vese [14] proposed the extension to vector case, where the model can detect edges present in at least one of the channels and not necessarily in all channels. By using contour information from all the components of the image vector, a highly detailed contour may be found which is superior to contours generated from a single dimension of the image vector.

The concepts of geodesic contours model and edgeless active contours model are conceptually different. The geodesic contours mechanism relies on gradients present in the image or image features and the edgeless approach considers the image to be a piecewise constant function. The two approaches were integrated into a unified algorithm in the work of Kimmel [28]. At first, the arclength term in [15] was replaced by the more general weighted arclength term, also known in the geodesic active contour functional [10] as a data sensitive regularization term. This yields better results in most cases and simplifies the regularization used by Chan and Vese [15]. Second, the alignment term as part of other driving forces of an active contour was added. The motivation is the fact, that in many cases the gradient direction is a good estimator for the orientation of the edge contour. The inner product gets high values if the curve normal aligns with the image gradient direction. This measure also uses the gradient magnitude as an edge indicator. The above combined approach which accounts for both the gradients between regions and region's homogeneity, may produce better results for various images.

Following the model [9] developed for grey level images, Sagiv, Sochen and Zeevi [50] generalized it to texture segmentation model. The energy functional to be minimized can be seen as a natural extension of the functional in [28], where the term which accounts for the arc-length of the curve is replaced by the geodesic length of the curve, weighted by the gradient information. The main novelty of this model is its representation of texture parameters as the embedding of a Riemannian surface in a higher dimensional space. This representation enables the definition of a Riemannian structure and its implementation in the definition of a texture edge indicator, which is subsequently used in a geodesic active contour algorithm for segmentation.

2.4 Geometric framework in image processing

Texture and color play important role in the understanding process of many images. Their study starts from the image formation on the retina and ends up with its interpretation by higher brain

functions. The geometric framework introduces the set of tools for enhancing while preserving either the multi-channel edges or their texture features. The images are treated as manifolds (surfaces in the $2D$ case) embedded in a higher dimensional space-feature manifold. This geometric interpretation leads to a simple way for image processing in grey level, color, video (i.e. dynamic images), volumetric medical data and texture. Moreover, it leads to schemes which are feature-preserving and hence suitable for segmentation tasks.

The manifold metric is a free "parameter" of the framework. Different choices lead to different schemes, as was shown by Sochen et. al. in [51]. For the induced metric the norm becomes simply the area or the volume of the image manifold, and the flow is towards minimal surface solution. Minimizing the area over the space of mappings, Sochen et. al. in [52] obtained the area minimization direction given by applying the second order differential operator of Beltrami on the feature coordinates, which in fact is a generalization of the Laplacian associated with flat spaces. In fact, for the grey level case the above operator describes the mean curvature flow of the image surface divided by the determinant of induced metric. The extension of the above approach for non-Euclidean embedding space was treated by Sochen and Zeevi in [53]. For $3D$ medical images and grey level video Kimmel et. al. [33] showed that in obtained Beltrami flow the determinant of induced metric may be considered as a generalized form of an edge indicator.

Another example of induced metric may be taken from [49]: Sagiv, Sochen and Zeevi proposed a novel approach for texture segmentation, which enables to use the geodesic active contours mechanism in the Gabor spatial feature space of images, by generalizing the edge indicator function. A Gabor-transformed image is represented as a $2D$ manifold with coordinates (x, y) embedded in the $6D$ feature space $(x, y, R(x, y), J(x, y), \theta(x, y), \sigma(x, y))$, where R and J are the real and imaginary parts of the Gabor transform, and θ and σ are the direction and scale for which a maximal value has been obtained. Alternatively, the transformed image may be represented as a $4D$ manifold with coordinates (x, y, θ, σ) embedded in the $6D$ feature space $(x, y, \theta, \sigma, R(x, y, \theta, \sigma), J(x, y, \theta, \sigma))$. The main difference between the two approaches is whether θ and σ are considered to be local or feature coordinates. In both cases the resulting stopping (edge indicator) function is the inverse of the determinant of the induced metric.

A novel and powerful approach has emerged from the mathematical theory of approximation of functionals via Γ-convergence. The idea is to approximate the functional by a different, parameter-dependent, functional that is expected to be more regular. The parameter controls the degree of approximation, such that the approximating functional approaches the original one in the limit, while the parameter approaches zero. The idea is that minimizers of approximating functional approximate the minimizer of original one, while enjoying greater regularity.

The above technique, merged with the Beltrami framework [51] was originally presented in [32]. Two different approaches are possible in the treatment of the segmentation function. First, the edge

indicator function E is defined on the image surface manifold $\{x, y, I(x, y)\}$. Here the Polyakov action [44] was used as an adaptive smoothing metric for both the color coordinates and the segmenting function. In the second approach the edge indicator function is defined over the image plane. The metric is as before, the induced metric, but this time the Polyakov action is used only for the feature coordinates. In the above work the generalized Γ-convergence technique was applied to Mumford-Shah functional with reference to two aspects: multi-channel images were analyzed and the $\mathbb{L}_2$ measure was replaced by the Polyakov action.

In [5] and [6], Brook, Kimmel and Sochen generalized the Mumford-Shah functional to color images, using the "image as a manifold" interpretation, while the length term remained the same. Assuming that the solution belongs to space of special functions with bounded variation (SBV), they suggested the generalized functional and the functionals series, Γ-converging to the above functional.

Chapter 3

Measure-based metric functions for Mumford-Shah functional

3.1 Preliminary analysis

Mumford and Shah [42] proposed the functional

$$F_{MS}(u,K) = \int\limits_{B\setminus K} |\nabla u|^2 + |K| + \alpha \int\limits_{B} |u-g|^2, \tag{3.1}$$

where $K \subset B$ is a one dimensional set which represents the edges, and $|K|$ is the length of this set (understood as the one-dimensional Hausdorff measure of K). A minimizer $\{K, u\}$ of F_{MS} thus produces the required image u as well as the edges K.

We propose an alternative functional. The concept behind this alternative is an object which is independent of parameterization, i.e a geometric functional which may be given in terms of the geometry of surfaces representing the data and image in the feature space (see below). Considering the image u as a two-dimensional surface, we shall replace the first two terms of (3.1) by the area of this surface. This allows sharp discontinuities (edges) of the image in the form of surface folding. This idea is not new (see, e.g, [51]). However, to make the third term of (3.1) fit the geometrical description, we must replace it by another metric $\mathcal{D}(g,u)$ representing the distance between the two surfaces.

To motivate this change, we point out that the term $\int |u-g|^2$ connects locally the image $u(b)$ for each $b \in B$ to the data g at *the same* point b. Thus, this term implicitly assumes that there is a one-to-one correspondence between points of the data to points of the image. It is well known that this is never the case, since defocusing and mapping errors imply that each pixel point is influenced by a domain rather than by a single point of the data.

Moreover, the proposed change allows us to replace the deterministic data g by a random one.

For this we replace $g(b)$ by a measure $\mu_b(dy)db$ where y is a parameterization of the feature fiber (see below). Further, to different pixels there may correspond different amount of data, so $\int \mu_b(dy) \leq 1$ for each pixel b where strong inequality and even zero value may not be excluded for some of the pixels.

So, our objective is, essentially, to define a metric $\mathcal{D}(\mu, U)$ between the data measure μ and the image surface U.

Before proceeding, let us define the feature space. Let $\mathbf{Y}$ be a set representing the possible data at a single pixel $b \in B$. It may be considered as a real number (the brightness) or a vector (if several color channels are present or, for example, a Gabor-wavelet filter, see [31]). We shall assume here that $\mathbf{Y} = \mathbb{R}^m$. The feature space is, then, defined as the cylinder $E := B \times \mathbf{Y} \subset \mathbb{R}^{m+2}$. The data g is represented as a measure μ supported in E.

To define an image u, let us consider the two-dimensional unit disc $S : \{|s| < 1\}$ diffeomorphic to B. We represent the image as a mapping $U : S \to E$ such that $U(\partial S) \subset \partial B \times \mathbf{Y}$ is projected onto ∂B. Coordinates in the feature space are described by $z := \{b, y\}$.

The feature space E is endowed with a metric

$$dz^2 = \gamma db^2 + dy^2 := \sum_{i,j=1}^{m+2} h_{i,j} dz_i dz_j,$$

where $db^2 = db_1^2 + db_2^2$, $dy^2 = \sum_{i=1}^{m} dy_i^2$ are the Euclidean metric in $\mathbb{R}^2$ and $\mathbb{R}^m$, respectively, while $\gamma > 0$ represents the relation between the geometric (pixel-domain) and feature metrics. Here

$$h_{i,j} = \begin{cases} \gamma, & 1 \leq i = j \leq 2 \\ 1, & 3 \leq i = j \leq m+2 \\ 0, & i \neq j \end{cases} .$$

With this setting, the embedding $U(S)$ into E is endowed with the induced metric

$$\Gamma_{i,j}(U) = U_{s_i}^T h U_{s_j} \quad \text{for} \quad i, j = 1, 2.$$

Setting $|\Gamma(U)| := \left|\Gamma_{1,1}(U)\Gamma_{2,2}(U) - \Gamma_{1,2}^2(U)\right|^{1/2}$, the surface area of $U(S)$ is

$$A(U) = \int_S |\Gamma(U)(s)| ds. \tag{3.2}$$

Let us now replace the first two terms of (3.1) by (3.2), and the last term by a distance $\mathcal{D}(\mu, U)$ between the embedded surface $U(S)$ and the data measure μ:

$$F(U) = A(U) + \alpha \mathcal{D}^2(\mu, U). \tag{3.3}$$

By our convention, the distance $\mathcal{D}$ should only depend on the image $U(S)$ and not on a particular parameterization. With this assumption, we replace $A(U)$ by the quadratic form

$$\widetilde{A}(U) = \frac{1}{2} \int_S Tr\left(\nabla U(s)^T h \nabla U(s)\right) ds, \tag{3.4}$$

where $Tr(\cdot)$ is a trace of a given matrix. Note that

$$\widetilde{A}(U) \geq A(U), \tag{3.5}$$

which reduces to an equality if the parameterization of U is conformal, namely

$$\Gamma_{1,1}(U) = \Gamma_{2,2}(U), \quad \Gamma_{1,2}(U) = \Gamma_{2,1}(U) = 0.$$

We replace $F(U)$ by

$$\widetilde{F}(U) = \widetilde{A}(U) + \alpha \mathcal{D}^2(\mu, U) \tag{3.6}$$

and obtain by (3.5)

$$\widetilde{F}(U) \geq F(U), \tag{3.7}$$

and equality if the embedded U is a conformal mapping from S to E.

For parametric representation of an image used in this section we cannot exclude the possibility that the optimal image is folded over the pixel space B. If this is the case, then some pixels of the image over B may have multiple values.

In continuation, this chapter is organized as follows. In Section 3.2 we introduce the distance function $D(\mu, \nu)$ and the associated metric $\mathcal{D}(\mu, U)$. The proposed definition of $\mathcal{D}$ has the following advantages: (i) it is lower semi-continuous in a proper topology, thus guarantees the existence of a minimizer, (ii) it can be relaxed into a form which is convenient for numerical implementation. In Section 3.3 we study the functionals F and $\widetilde{F}$ over suitable domains. In particular, we show that a minimizer of $\widetilde{F}$ exists and is a conformal mapping, hence it is also a minimizer of F. In Section 3.4 we introduce the relaxation of D into D_β for $\beta > 0$ such that $\lim_{\beta \to 0} D_\beta = D$ is in the sense of Γ-convergence. This, in particular, will enable us in Section 3.5 to relax $\widetilde{F}$ into $\widetilde{F}_\beta$, whose Γ-limit at $\beta \to 0$ coincides with $\widetilde{F}$.

3.2 The distance function and parameter dependent metric

Our first object is to define a distance between two positive measures μ and ν supported in E, where ν represents the image and μ the data. We shall define this distance in terms of the supports of the measures, and show that it is lower semi-continuous with respect to the C^* topology of measures. Then we shall write the distance where ν is given in terms of the embedding U and show that it

is weakly lower semi-continuous with respect to the $\mathbb{H}^1$ norm, which is the natural one for (3.4). Finally, we prove the existence of minimizers to both F and $\widetilde{F}$ in a proper setting, and show that the sets of minimizers of both functionals coincide.

To motivate the discussion, recall that the Hausdorff distance d_H between two closed sets $A, B \subset E$ can be written as:

$$d_H(A,B) = \sqrt{\sup_{v\in B}\inf_{u\in A}|u-v|^2} + \sqrt{\sup_{u\in A}\inf_{v\in B}|u-v|^2}.$$

It is tempting to define the distance between a pair of measures by

$$D(\mu,\nu) = d_H(supp(\mu), supp(\nu)).$$

However, it turns out that, by this definition, D is not C^* lower semi-continuous with respect to ν. So we define

$$d(A,B) = \sqrt{\sup_{v\in B}\inf_{u\in A}|u-v|^2}$$

and

$$D(\mu,\nu) = d(supp(\mu), supp(\nu)). \tag{3.8}$$

We shall see below that D is C^* lower semi-continuous with respect to ν (but not with respect to μ). Indeed, the data μ is fixed, so we have to care only about the dependence on ν, which represents the image on which we minimize.

Given a continuous function f on E, let

$$\sup_{\nu}(f) = \sup_{v\in supp(\nu)} f(v). \tag{3.9}$$

Similarly

$$\inf_{\mu}(f) = \inf_{u\in supp(\mu)} f(u). \tag{3.10}$$

To show the next fact, we use the Portmanteau Theorem ([4], Section 1.2, Theorem 2.1), which, in our present notation, reads respectively in a following manner:

Theorem 3.1 *Let ν_n, ν be the measures in C^*. The following conditions are equivalent:*

- $\nu_n \rightharpoonup \nu$ *in* C^*.
- $\lim_{n\to\infty}\int f d\nu_n = \int f d\nu$ *for all bounded, uniformly continuous real* f.
- $\limsup_{n\to\infty}\nu_n(F) \le \nu(F)$ *for all closed* F.
- $\liminf_{n\to\infty}\nu_n(G) \ge \nu(G)$ *for all open* G.

Proof: See [4].

■

We now claim:

Lemma 3.1 *For fixed, continuous f, $\sup_{\nu}(f)$ is lower semi-continuous under the topology of weak C^* convergence.*

Proof: Due to compactness of C^* we may isolate a weakly converging subsequence ν_n to some measure ν. Denote also by $\overline{v}$ a point in which f attains its supremum within $supp(\nu)$ (because $supp(\nu)$ is closed). Let us show that for every $\varepsilon > 0$ there exists $N \geq 0$, s.t. for $n \geq N$ $supp(\nu_n) \bigcap B_\varepsilon(\overline{v}) \neq \emptyset$ ($B_\varepsilon(\overline{v})$ is an open ball of radius ε centered at $\overline{v}$). Suppose on the contrary, that there exists $\varepsilon > 0$ s.t. for every N there exists sequence $n_k \geq N$, s.t. $supp(\nu_{n_k}) \bigcap B_\varepsilon(\overline{v}) = \emptyset$. It follows from here that

$$\liminf_{n\to\infty} \nu_n(B_\varepsilon(\overline{v})) = \lim_{k\to\infty} \nu_{n_k}(B_\varepsilon(\overline{v})) = 0.$$

On the other hand, $supp(\nu) \bigcap B_\varepsilon(\overline{v}) \neq \emptyset$ ($\overline{v} \in supp(\nu)$), thus $\nu(B_\varepsilon(\overline{v})) > 0$. Hence,

$$\liminf_{n\to\infty} \nu_n(B_\varepsilon(\overline{v})) < \nu(B_\varepsilon(\overline{v}))$$

and this is a contradiction to the Theorem 3.1: it should be $\liminf_{n\to\infty} \nu_n(B_\varepsilon(\overline{v})) \geq \nu(B_\varepsilon(\overline{v}))$.

Recall the original assumption: for every $\varepsilon > 0$ there exists $N \geq 0$, s.t. for $n \geq N$ $supp(\nu_n) \bigcap B_\varepsilon(\overline{v}) \neq \emptyset$. Hence, there exists the sequence $v_n \in supp(\nu_n)$, s.t. for $n \geq N$ $|v_n - \overline{v}| < \varepsilon$, i.e. $\lim_{n\to\infty} v_n = \overline{v}$. Taking into account that $\sup_{v\in supp(\nu)} f(v) \geq f(v_n)$, we obtain

$$\liminf_{n\to\infty} \sup_{\nu_n}(f) = \liminf_{n\to\infty} \sup_{v\in supp(\nu_n)} f(v) \geq \liminf_{n\to\infty} f(v_n) = f(\overline{v}) = \sup_{v\in supp(\nu)} f(v) = \sup_{\nu}(f).$$

■

Assume from now, that μ is compactly supported in E. Setting $f(v) = \inf_{u\in supp(\mu)} |u - v|^2$, we obtain as a corollary:

Corollary 3.1 *For fixed, compactly supported μ, $D(\mu, \nu)$ is lower semi-continuous with respect to ν under the topology of weak C^* convergence.*

Proof: Follows immediately from Lemma 3.1 and the fact that $f(v)$ is continuous.

■

We now replace the measure ν by a mapping U. Let l be a measure on the parameter space S (say, the uniform Lebesgue measure). We denote the measure ν associated with an embedding $U \in \mathbb{H}^1(S, E)$ by ν_U and define it by the pullback

$$\nu_U(\sigma) = l\left(U^{-1}(\sigma \cap U(S))\right)$$

for every Borel measurable set $\sigma \subset E$. In terms of its action on a test function $\phi \in C_0(E)$ the above measure is defined in the following way:

$$\langle \nu_U, \phi \rangle = \int_S \phi(U(s)) ds. \tag{3.11}$$

Next we establish the useful property of the measure ν_U.

Proposition 3.1 *If U_n converges weakly to U in $\mathbb{H}^1(S,E)$, then ν_{U_n} converges weakly in C^* to ν_U.*

Proof: The proposition is equivalent to

$$\lim_{n\to\infty} \int_S \phi(U_n(s)) ds = \int_S \phi(U(s)) ds \tag{3.12}$$

for any $\phi \in C_0(E)$. Any function in $C_0(E)$ may be approximated by $C_0^1(E)$ function, thus it is enough to show (3.12) for $\phi \in C_0^1(E)$. It is obvious that

$$\begin{aligned} \left| \int_S \phi(U_n(s)) ds - \int_S \phi(U(s)) ds \right| &\leq \int_S |\phi(U_n(s)) - \phi(U(s))| \, ds \leq \max_E |\nabla \phi| \int_S |U_n(s) - U(s)| ds \\ &\leq \max_E |\nabla \phi| \cdot \sqrt{|S|} \cdot \sqrt{\|U_n - U\|_{\mathbb{L}^2}}. \end{aligned}$$

From the compact embedding of $\mathbb{H}^1$ into $\mathbb{L}^2$ we obtain that the right-hand part tends to 0 when $n \to \infty$.

■

We conclude immediately that (3.11) extends to any $U \in \mathbb{H}^1(S,E)$ and define

$$\mathcal{D}(\mu, U) := D(\mu, \nu_U).$$

We now claim:

Corollary 3.2 *The metric $\mathcal{D}(\mu, U)$ is weakly lower semi-continuous with respect to U in the $\mathbb{H}^1$ topology.*

Proof: Follows immediately from Corollary 3.1 and Proposition 3.1.

■

3.3 Existence of the minimizer of functional $\widetilde{F}$

The natural domain for the functional $\widetilde{F}$ (3.6) is the space $\mathbb{H}^1(S,E)$. However, a minimizer of $\widetilde{F}$ over this set is trivial, since for the constant map $U = z_0$ with $z_0 \in supp(\mu)$ we certainly have $\widetilde{F}(U) = 0$. To avoid this, we need to implement a proper boundary condition for $\mathbb{H}^1(S,E)$.

By the interpretation of this functional, a natural requirement is that U should be a mapping of S onto a disc-like set in E whose boundary $U(C_S)$, $C_S := \partial S$ is the "frame" of the image, given by a Jordan curve on the cylinder $\mathbf{C} := \partial B \times \mathbf{Y}$. However, this curve must not be specified a-priori, since we assume no a-priori knowledge on the boundary of this image. Referring to the classical theory of minimal surfaces, this suggests to consider the cylinder $\mathbf{C}$ as a free boundary and to limit the domain of $\widetilde{F}$ to a subset of $\mathbb{H}^1(S,E)$ composed of mappings whose boundary values are contained in the identity of the homotopy class π_1 of the cylinder $\mathbf{C}$. In fact, such a boundary class can be defined in a generalized sense for $\mathbb{H}^1$ (see [18], ch.5). However, this class is not closed under the weak topology of $\mathbb{H}^1$, and this presents a major difficulty for the proof that a limit of minimizing sequence preserves the boundary condition (see [18], ch.5.1, Theorem 4 for a counter-example).

Fortunately, the free boundary in our case is the cylinder and we may use separation of variables to circumvent this problem. Let $U : S \to E$ be written in the form $U := \{U^B, U^{\mathbf{Y}}\}$ where $U^B : S \to B$ and $U^{\mathbf{Y}} : S \to \mathbf{Y}$. Consider the functional (3.6) written as:

$$\widetilde{F}(U^B, U^{\mathbf{Y}}) = \frac{1}{2} \int_S \left[\gamma Tr\left(\nabla U^B(s)^T \nabla U^B(s) \right) + Tr\left(\nabla U^{\mathbf{Y}}(s)^T \nabla U^{\mathbf{Y}}(s) \right) \right] ds + \alpha \mathcal{D}^2(\mu, U). \quad (3.13)$$

This representation allows us to define the domain of $U^{\mathbf{Y}}$ as $\mathbb{H}^1(S,\mathbf{Y})$ without any boundary condition:

$$\mathcal{DOM}_{\mathbf{Y}} := \{U^{\mathbf{Y}} \in \mathbb{H}^1(S,\mathbf{Y})\}.$$

The mapping U^B, on the other hand, must map S onto B such that $U^B|_{C_S} : C_S \to C_B := \partial B$ is a homeomorphism. This corresponds to treating U_B as in a Plateau problem with *non-free* boundary condition. Following the general wisdom for this type of problems, we need the technical 3-point condition (see [18], p. 235) and set:

$$\begin{aligned} \mathcal{DOM}_B \;:=\; & \{U^B \in \mathbb{H}^1(S,B) \cap C^0(C_S, C_B) \mid \; U^B|_{C_S} : C_S \to C_B \textit{ is a homeomorphism such} \\ & \textit{that } \; U^B(s_i) = \zeta_i \textit{ for } i = 1,2,3, \textit{ where } s_1, s_2, s_3 \textit{ are three distinct points on} \\ & C_S, \textit{ while } \zeta_1, \zeta_2, \zeta_3 \textit{ are three distinct points on } C_B \textit{ with the same ordering}\} \end{aligned}$$

To show the next fact, we use the $\varepsilon-$conformal mapping theorem of Morrey ([39], pp. 814-815) which, in our present notation, reads:

Theorem 3.2 *For a mapping $U \in \mathbb{H}^1(S,E) \cap C^0(C_S, C)$, for every $\varepsilon > 0$ there exists a C^2 homeomorphism τ_ε from $\overline{S}$ onto itself which satisfies*

$$Z_\varepsilon := U \circ \tau_\varepsilon \in \mathbb{H}^1(S,E) \cap C^0(C_S, C)$$

and

$$\widetilde{A}(Z_\varepsilon) \leq A(U) + \varepsilon,$$

where A and $\widetilde{A}$ are defined as in (3.2) and (3.4) respectively.

Proof: See [39].

■

The following theorem is the main goal of this section.

Theorem 3.3 *If μ has a compact support in E then a minimizer of $\widetilde{F}$ is attained in the domain $\mathcal{DOM} := \mathcal{DOM}_B \times \mathcal{DOM}_{\mathbf{Y}}$. Moreover, any minimizer $\overline{U}$ is a minimizer of F (3.3) as well.*

Proof: Let $U_n := \left\{U_n^B, U_n^{\mathbf{Y}}\right\}$ be a minimizing sequence of $\widetilde{F}$. By the compactness of $supp(\mu)$ and the definition of $\mathcal{D}$ we obtain that U_n are uniformly bounded in $\mathbb{L}^\infty(S,E)$. Since the Dirichlet term in $\widetilde{F}$ is bounded as well, we conclude that U_n is a bounded sequence in $\mathbb{H}^1(S,E)$. Therefore (see [35], Section 2.18), there is a weak limit $\overline{U} := \left\{\overline{U}^B, \overline{U}^{\mathbf{Y}}\right\} \in \mathbb{H}^1(S,E)$. Thus, we only have to show that $\overline{U}^B \in \mathcal{DOM}_B$. This follows by the standard proof of the Plateau problem (see, e.g., [18], ch.4): We apply the Courant-Lebesgue Lemma and the three-point condition to obtain the equicontinuity of the traces $U_n^B|_{C_S}$. The Arzela-Ascoli theorem implies the convergence of the traces to a continuous, weakly monotonic function $\phi : C_S \to C_B$ which preserves the three point condition. On the other hand, the traces $U_n^B|_{C_S}$ converge in $\mathbb{L}^2(C_S, C_B)$ by compact embedding of the space of traces $\mathbb{H}^{1/2}(C_S, C_B)$ in $\mathbb{L}^2(C_S, C_B)$. This $\mathbb{L}^2$ limit of the traces coincides with ϕ. This implies that $\overline{U}^B \in \mathcal{DOM}_B$.

Denote $Z_\varepsilon = \overline{U} \circ \tau_\varepsilon$, where τ_ε is a C^2 homeomorphism. Taking an advantage of the fact that $\overline{U}$ is the minimizer of $\widetilde{F}$ and using Theorem 3.2, we obtain

$$\begin{aligned}\widetilde{F}(\overline{U}) &\leq \widetilde{F}(Z_\varepsilon) = \widetilde{A}(Z_\varepsilon) + \alpha\mathcal{D}^2(\mu, Z_\varepsilon) = \widetilde{A}(Z_\varepsilon) + \alpha D^2(\mu, \nu_{Z_\varepsilon}) = \widetilde{A}(Z_\varepsilon) + \alpha D^2(\mu, \nu_{\overline{U}}) \\ &\leq A(\overline{U}) + \varepsilon + \alpha D^2(\mu, \nu_{\overline{U}}) = F(\overline{U}) + \varepsilon\end{aligned}$$

for every $\varepsilon > 0$. Passing to the limit $\varepsilon \to 0$, together with inequality $F(\overline{U}) \leq \widetilde{F}(\overline{U})$ we complete the proof of the theorem.

■

3.4 Relaxation of the distance function

In this section we are going to use a novel and powerful approach that has appeared from the mathematical theory of approximation of functionals via Γ-convergence. The idea is to approximate the functional with the lack of regularity by a series of different, parameter dependent functionals, that are expected to be more regular. In addition, we expect from convergence of above functionals to imply the convergence of minimizers. Γ-convergence has this nice property.

Definition 3.1 *Let (S,d) be a separable metric space and let $F_n : S \to [0,+\infty], \quad n = 1,2,\ldots$ be a sequence of functions. The sequence F_n is said to Γ-converge to $F : S \to [0,+\infty]$ if the following two*

conditions hold for all $f \in S$:

$$\begin{aligned} \forall f_n \to f \qquad & \liminf_{n\to\infty} F_n(f_n) \geq F(f), \\ \textit{and} \quad \exists f_n \to f \qquad & \limsup_{n\to\infty} F_n(f_n) \leq F(f). \end{aligned}$$

The following theorem (see [2], Proposition 2.1) characterizes the main properties of Γ-convergence.

Theorem 3.4 *Assume that $\{F_n\}$ Γ-converges to F. Then, the following statements hold:*

(i) Let $t_n \downarrow 0$, then every cluster point of the sequence of sets

$$\{f \in S : F_n(f) \leq \inf_S F_n + t_n\}$$

minimizes G.

(ii) Assume that the functionals F_n are lower semi-continuous and, for every $t \in [0, \infty)$, there exists a compact set $C_t \subset S$ such that for all n $\{f \in S : F_n(f) \leq t\} \subset C_t$. Then the functionals F_n have minimizers in S, and any sequence of minimizers of F_n admits subsequences converging to some minimizer of F.

Proof: See [2].

■

The parameter n controls the degree of approximation, such that the approximating functional F_n equals to original one F in the limit, while the parameter n goes to infinity. The approach is that minimizers of approximating functionals approximate the minimizer of original one, while enjoying greater regularity. Whenever the limit F exists, it is unique and lower semi-continuous. We can extend this definition to families of functionals depending on continuous parameter $\beta \downarrow 0$, requiring convergence of F_{β_n} to F on every sequence $\beta_n \downarrow 0$.

It is important to notice that Γ-limit depends on what kind of convergence we have on S.

First let define a relaxation of the $\sup_\nu$ and $\inf_\mu$ functionals (3.9) and (3.10). Given $\beta > 0$, set

$$\overset{\beta}{\sup_\nu}(f) = \beta \ln \left[\int_E e^{f(v)/\beta} \nu(dv) \right],$$

$$\overset{\beta}{\inf_\mu}(f) = -\beta \ln \left[\int_E e^{-f(u)/\beta} \mu(du) \right].$$

Lemma 3.2 *For a fixed continuous function f in E and fixed Borel measures μ and ν supported in E:*

$$\lim_{\beta\to 0} \overset{\beta}{\sup_\nu}(f) = \sup_\nu(f), \tag{3.14}$$

$$\lim_{\beta\to 0}\overset{\beta}{\inf_{\mu}}(f)=\inf_{\mu}(f). \tag{3.15}$$

Moreover, if ν_n converges C^* to ν and $\beta_n\to 0$ from above ($\beta_n\geq 0$), then

$$\liminf_{n\to\infty}\overset{\beta_n}{\sup_{\nu_n}}(f)\geq\sup_{\nu}(f). \tag{3.16}$$

Proof: Let show first (3.14). Let $v_0\in supp(\nu)$ s.t. $\sup\limits_{\nu} f=f(v_0)$. Then

$$\overset{\beta}{\sup_{\nu}}(f)=f(v_0)+\beta\ln\left[\int_E e^{(f(v)-f(v_0))/\beta}\nu(dv)\right].$$

Evidently

$$\lim_{\beta\to 0}\beta\ln\left[\int_E e^{(f(v)-f(v_0))/\beta}\nu(dv)\right]\leq 0. \tag{3.17}$$

To show the reverse inequality in (3.17), let $B_\varepsilon(v_0)$ be a ball of radius ε centered at v_0. By continuity of f we have $\inf\limits_{B_\varepsilon(v_0)} f\geq f(v_0)-d(\varepsilon)$, where $\lim\limits_{\varepsilon\to 0}d(\varepsilon)=0$. Moreover, $\nu(B_\varepsilon(v_0))>0$ since $v_0\in supp(\nu)$. Then

$$\beta\ln\left[\int_E e^{(f(v)-f(v_0))/\beta}\nu(dv)\right]\geq\beta\ln\left[\nu(B_\varepsilon(v_0))e^{-d(\varepsilon)/\beta}\right]=-d(\varepsilon)+\beta\ln\left[\nu(B_\varepsilon(v_0))\right]. \tag{3.18}$$

Taking now the limit $\beta\to 0$, we obtain that the limit in (3.18) is no smaller than $-d(\varepsilon)$. Since ε (hence $d(\varepsilon)$) can be chosen arbitrary small, we have the equality in (3.17).

The proof of (3.15) is analogous. To prove (3.16), we proceed as in (3.18), where $\nu(B_\varepsilon(v_0))$ is replaced by $\nu_n(B_\varepsilon(v_0))$, and note that $\lim\limits_{n\to\infty}\nu_n(B_\varepsilon(v_0))=\nu(B_\varepsilon(v_0))$.

■

Let define now

$$q_\beta(v):=\overset{\beta}{\inf_{\mu}}(|u-v|^2)=-\beta\ln\left[\int_E e^{-|u-v|^2/\beta}\mu(du)\right] \tag{3.19}$$

and

$$\overset{\beta}{\sup_{\nu}}(q_\beta)=\beta\ln\left[\int_E e^{q_\beta(v)/\beta}\nu(dv)\right]. \tag{3.20}$$

Now, inserting (3.19) into (3.20) we define the relaxation D_β of D as:

$$D_\beta(\mu,\nu)=\sqrt{\beta\ln\left[\int_E\frac{\nu(dv)}{\int_E e^{-|u-v|^2/\beta}\mu(du)}\right]}.$$

Using Lemma 3.2 we obtain:

Corollary 3.3 *$D(\mu,\cdot)$ is the $\Gamma-$limit of $D_\beta(\mu,\cdot)$ as $\beta\to 0$ from above, if considered as a functional of the second argument. That is:*

$$\lim_{\beta\to 0;\beta>0} D_\beta(\mu,\nu) = D(\mu,\nu),$$

$$\liminf_{n\to\infty} D_{\beta_n}(\mu,\nu_n) \geq D(\mu,\nu),$$

where $\beta_n\to 0$ from above and $\nu_n \rightharpoonup \nu$ in C^*.

Proof: Follows immediately from Lemma 3.2 and the fact that $q_\beta(v)$ is continuous.

■

Remark: Corollary 3.3 still holds if we assume $\beta\to 0$ from *below*. However, in this case, the $\Gamma-$ limit is not to D as defined by (3.8), but to $D_1(\mu,\nu) = d_1(supp(\mu), supp(\nu))$ where

$$d_1(A,B) = \sqrt{\inf_{v\in A}\sup_{u\in B}|u-v|^2}.$$

In particular,

$$\lim_{\beta\to 0;\ \beta>0} D_\beta(\mu,\nu) \neq \lim_{\beta\to 0;\ \beta<0} D_\beta(\mu,\nu)$$

in general(!). It is easy to see that D_1 is also C^* lower semi-continuous with respect to ν. However, it is not a good distance function. Note that $D_1(\mu,\nu) > 0$ if $supp(\nu)$ is not a single point and $supp(\nu)\subset supp(\mu)$, while $D(\mu,\nu) = 0$ under the above conditions.

3.5 Relaxation of the original functional

Here we introduce an implementation of the relaxed distance function D_β and define a corresponding version of the metric $\mathcal{D}_\beta(\mu,U)$, $U\in \mathbb{H}^1(S,E)$. This enables us to introduce the relaxation of $\widetilde{F}$ (3.6) by

$$\widetilde{F}_\beta(U) = \frac{1}{2}\int_S Tr\left(\nabla U(s)^T h \nabla U(s)\right) ds + \alpha\mathcal{D}^2_\beta(\mu,U). \tag{3.21}$$

We set the $\beta-$distance between a mapping $U: S\to E$ to a measure μ as

$$\mathcal{D}_\beta(\mu,U) := D_\beta\left(\mu,\nu_U\right) = \sqrt{\beta\ln\left[\int_S \frac{ds}{\int_E e^{-|U(s)-z|^2/\beta}\mu(dz)}\right]}. \tag{3.22}$$

Thus

$$\mathcal{D}^2_\beta(\mu,U) = \beta\ln\left[\int_S \Xi_\beta(U)(s)ds\right],$$

where

$$\Xi_\beta(U)(s) = \frac{1}{\int\limits_E e^{-|U(s)-z|^2/\beta}\mu(dz)}.$$

We now claim:

Corollary 3.4 *$\mathcal{D}(\mu,\cdot)$ is the $\Gamma-$limit of $\mathcal{D}_\beta(\mu,\cdot)$ as $\beta\to 0$ from above.*

Proof: Follows immediately from Corollary 3.3 and Proposition 3.1.

■

We may now write explicitly the functional (3.21):

$$\widetilde{F}_\beta(U) = \frac{1}{2}\int\limits_S Tr\left(\nabla U(s)^T h \nabla U(s)\right) ds + \alpha\beta \ln\left[\int\limits_S \Xi_\beta(U)(s)ds\right]. \tag{3.23}$$

The following theorem is the main goal of this section.

Theorem 3.5 *For any $\beta>0$, the functional (3.23) attains a minimizer $U_\beta\in\mathcal{DOM}$. The sequence of minimizers $\{U_\beta\}$ is bounded in $\mathbb{H}^1(S,E)$. Any weak limit U in $\mathbb{H}^1$, s.t.*

$$U_\beta \overset{\beta\to 0}{\rightharpoonup} U\in\mathcal{DOM}, \tag{3.24}$$

is a minimizer of F (3.3) and $\widetilde{F}$ and is a conformal embedding of S into E.

Proof: Let U_n be a minimizing sequence of $\widetilde{F}_\beta(U)$ for fixed $\beta>0$. The proof follows along the line of Theorem 3.3, except for the absence of the a-priori $\mathbb{L}^\infty$ estimate of U_n. However, using

$$|U_n(s)-z|^2 \geq (1-\varepsilon)|U_n(s)|^2 + (1-\varepsilon^{-1})|z|^2,$$

for $\varepsilon>0$ we have

$$\Xi_\beta(U_n)(s) \geq \frac{e^{(1-\varepsilon)|U_n(s)|^2/\beta}}{\int\limits_E e^{-(1-\varepsilon^{-1})|z|^2/\beta}\mu(dz)}.$$

By compactness of $supp(\mu)$ and the choice $\varepsilon=1/2$ we obtain an a-priori estimate

$$\int\limits_S e^{|U_n(s)|^2/2\beta} \leq \int\limits_S \Xi_\beta(U)(s)ds \cdot \int\limits_E e^{|z|^2/\beta}\mu(dz),$$

which is enough (since we actually need only $\mathbb{L}^2$ upper estimate for U_n).

The second part of the theorem follows from Corollary 3.4.

■

The functional $\widetilde{F}_\beta$ equipped with the relaxed distance $\mathcal{D}_\beta$ is no longer parameter independent. In particular, its minimizers are not conformal mappings. However, it is converging to parameter independent functional $\widetilde{F}$ as $\beta\to 0$.

Chapter 4

Parametric representation

4.1 Variational formulation and Euler-Lagrange equation for the relaxed functional

Let consider a parametric representation of the image U. In this case also $U(S) := \{U^B, U^{\mathbf{Y}}\}$, where U^B is the projection of U on the pixel space B and $U^{\mathbf{Y}}$ is the projection of U on the feature fiber $\mathbf{Y}$. We also separate the scale γ into γ_1 for the area term and γ_2 for the metric. For such formulation the area term of functional (3.23) is given as in (3.4):

$$\widetilde{A}(U) = \frac{1}{2}\int\limits_S Tr\left(\nabla U(s)^T h \nabla U(s)\right) ds = \frac{1}{2}\int\limits_S \left(\|U_{s_1}\|^2_{\gamma_1} + \|U_{s_2}\|^2_{\gamma_1}\right) ds,$$

where

$$\|V\|_\gamma = \sqrt{\gamma\left(V^1\right)^2 + \gamma\left(V^2\right)^2 + \sum_{k=3}^{m+2}\left(V^k\right)^2} = \sqrt{\gamma\left(V^B\right)^2 + \left(V^{\mathbf{Y}}\right)^2}.$$

Let now define the relaxed metric term $\mathcal{D}_\beta(\mu, U)$. The parameter dependent metric (3.22) attributes the measure ds to the image U, and it is written as

$$\mathcal{D}^2_{\beta,\gamma_2}(\mu, U) = \beta \ln\left[\int\limits_S \Xi_{\beta,\gamma_2}(U)(s) ds\right], \tag{4.1}$$

where

$$\Xi_{\beta,\gamma_2}(U)(s) = \frac{1}{\int\limits_E e^{-|U(s)-z|^2_{\gamma_2}/\beta}\mu(dz)}. \tag{4.2}$$

For a parametric formulation we represent the data $\mu(dz)$ as $\mu_s(dy)ds$. In addition, the following analysis is performed for deterministic data $G \in \mathbb{L}^\infty(S, E)$, thus $\Xi_{\beta,\gamma_2}(U)(s)$ in (4.2) is given as

$$\Xi_{\beta,\gamma_2}(U)(s) = \frac{1}{\int\limits_S e^{-|U(s)-G(s')|^2_{\gamma_2}/\beta} ds'}. \tag{4.3}$$

Finally, the corresponding functional is:

$$\mathbf{F}^{\beta}_{\gamma_1,\gamma_2}(U) = \frac{1}{2}\int\limits_S \left(\|U_{s_1}\|^2_{\gamma_1} + \|U_{s_2}\|^2_{\gamma_1}\right) ds + \alpha\beta \ln\left[\int\limits_S \Xi_{\beta,\gamma_2}(U)(s)ds\right], \tag{4.4}$$

where $\Xi_{\beta,\gamma_2}(U)(s)$ is given as in (4.3).

We aim to determine the condition for the first variation of the functional (4.4) to be 0. This is a necessary condition for U to be a minimizer. Let suppose that the boundary of B is given by

$$g(U^1, U^2) = 0. \tag{4.5}$$

Then, the minimum is the function U which solves the boundary value problem

$$\begin{cases} \Delta U^i - \dfrac{\gamma_2}{\gamma_1}\cdot\dfrac{2\alpha\left(\Xi_{\beta,\gamma_2}(U)(s)\right)^2}{\int\limits_S \Xi_{\beta,\gamma_2}(U)(s)ds}\int\limits_S e^{-|U(s)-G(s')|^2_{\gamma_2}/\beta}(U^i(s) - G^i(s'))ds' = 0, \quad i = 1,2 \\ \Delta U^i - \dfrac{2\alpha\left(\Xi_{\beta,\gamma_2}(U)(s)\right)^2}{\int\limits_S \Xi_{\beta,\gamma_2}(U)(s)ds}\int\limits_S e^{-|U(s)-G(s')|^2_{\gamma_2}/\beta}(U^i(s) - G^i(s'))ds' = 0, \quad i = 3,\ldots,m+2 \end{cases}. \tag{4.6}$$

In addition to (4.5),

$$g_{U_1}(U^1, U^2)\frac{\partial U^2}{\partial \vec{\nu}} - g_{U_2}(U^1, U^2)\frac{\partial U^1}{\partial \vec{\nu}} = 0 \quad \text{on } \partial S \tag{4.7}$$

is a second boundary condition for $U^i,\ i = 1,2$. Neumann boundary conditions

$$\frac{\partial U^i}{\partial \vec{\nu}} = 0 \quad \text{on } \partial S \tag{4.8}$$

are natural boundary conditions for $U^i,\ i = 3,\ldots,m+2$ respectively.

4.2 Interior regularity of the minimizer of relaxed functional

In Section 3.5 we have proved that for any $\beta > 0$, the functional (4.4) attains a minimizer $U_\beta \in \mathcal{DOM}$, i.e. the system (4.6) has a weak solution $U_\beta \in \mathbb{H}^1(S,E)$. We now address the question of regularity of such a minimizer. In fact, in this section we intend to show the infinite interior differentiability of the minimizer U_β.

Denote

$$f^i(U)(s) = \left(\Xi_{\beta,\gamma_2}(U)(s)\right)^2\int\limits_S e^{-|U(s)-G(s')|^2_{\gamma_2}/\beta}(U^i(s) - G^i(s'))ds', \quad i = 3,\ldots,m+2.$$

Proposition 4.1 *Let $U_\beta \in \mathbb{H}^1(S,E)$ be a minimizer of the functional*

$$\mathbf{F}^{\beta}_{\gamma_1,\gamma_2}(U) = \frac{1}{2}\int\limits_S \left(\|U_{s_1}\|^2_{\gamma_1} + \|U_{s_2}\|^2_{\gamma_1}\right) ds + \alpha\beta \ln\left[\int\limits_S \Xi_{\beta,\gamma_2}(U)(s)ds\right].$$

Then $U_\beta \in \mathbb{L}^\infty(S,E)$, if $G \in \mathbb{L}^\infty(S,E)$. Moreover, $\|U_\beta\|_\infty \le \|G\|_\infty$.

Proof: Suppose by contradiction that $U_\beta \notin \mathbb{L}^\infty(S,E)$. Denote $R = \|G\|_\infty$ and define the retraction

$$\tilde{U}_\beta(s) = \begin{cases} R\dfrac{U_\beta(s)}{|U_\beta(s)|}, & |U_\beta(s)| \geq R \\ U_\beta(s), & |U_\beta(s)| < R \end{cases}.$$

From the definition of retraction it is obvious that

$$\left\|\frac{\partial \tilde{U}_\beta}{\partial s_1}\right\|_{\gamma_1}^2 + \left\|\frac{\partial \tilde{U}_\beta}{\partial s_2}\right\|_{\gamma_1}^2 < \left\|\frac{\partial U_\beta}{\partial s_1}\right\|_{\gamma_1}^2 + \left\|\frac{\partial U_\beta}{\partial s_2}\right\|_{\gamma_1}^2 \tag{4.9}$$

and

$$|\tilde{U}(s) - G(s')| \leq |U(s) - G(s')|,$$

thus,

$$\Xi_{\beta,\gamma_2}(\tilde{U}_\beta)(s) < \Xi_{\beta,\gamma_2}(U_\beta)(s). \tag{4.10}$$

Taking together (4.9) and (4.10), we obtain that

$$\mathbf{F}^\beta_{\gamma_1,\gamma_2}(\tilde{U}_\beta) < \mathbf{F}^\beta_{\gamma_1,\gamma_2}(U_\beta),$$

in the contradiction with assumption that U_β is the minimizer of $\mathbf{F}^\beta_{\gamma_1,\gamma_2}(U)$.

■

We aim to determine now the upper estimates for $\Xi_\beta(U_\beta)$ and $\dfrac{1}{\int\limits_S \Xi_\beta(U_\beta)(s)ds}$, which will serve us in the following. Let assume, as in previous proposition, that $\left|G^i(s')\right| \leq R$ and consequently, $\left|U^i_\beta(s)\right| \leq R$. Using the fact that

$$|U_\beta(s) - G(s')|^2_{\gamma_2} \leq (|U_\beta(s)|_{\gamma_2} + |G(s')|_{\gamma_2})^2 \leq 4(2\gamma_2 + m)^2R^2,$$

we have

$$\Xi_\beta(U_\beta)(s) \leq \frac{1}{\int\limits_S e^{-4(2\gamma_2+m)^2R^2/\beta}ds'} = \frac{1}{\pi}e^{4(2\gamma_2+m)^2R^2/\beta}, \tag{4.11}$$

which in turn means that

$$\Xi_\beta(U_\beta) \in \mathbb{L}^\infty(S).$$

Taking into account the following inequality

$$|U_\beta(s) - G(s')|^2_{\gamma_2} \geq (1-\varepsilon)|U_\beta(s)|^2_{\gamma_2} + (1-\varepsilon^{-1})|G(s')|^2_{\gamma_2},$$

for $\varepsilon = \frac{1}{2}$ we obtain

$$\Xi_\beta(U_\beta)(s) \geq \frac{e^{|U_\beta(s)|^2_{\gamma_2}/2\beta}}{\int\limits_S e^{|G(s')|^2_{\gamma_2}/\beta}ds'} \geq \frac{1}{\int\limits_S e^{(2\gamma_2+m)^2R^2/\beta}ds'} = \frac{1}{\pi}e^{-(2\gamma_2+m)^2R^2/\beta},$$

and thus,

$$\frac{1}{\int\limits_S \Xi_\beta(U_\beta)(s)ds} \leq e^{(2\gamma_2+m)^2R^2/\beta}. \tag{4.12}$$

Corollary 4.1 *Let $U_\beta \in \mathbb{L}^\infty(S,E)$. Then $f^i(U_\beta) \in \mathbb{L}^\infty(S)$ for $i = 1,\ldots,m+2$.*

Proof: Follows immediately from (4.11), the fact that $e^{-|U(s)-G(s')|^2_{\gamma_2}/\beta} \leq 1$ and $|U^i(s)-G^i(s')| \leq 2R$.

■

Let consider now the following system of semi-linear elliptic equations of the second order

$$\Delta U^i = \begin{cases} \dfrac{\gamma_2}{\gamma_1} \cdot \dfrac{2\alpha}{\int\limits_S \Xi_{\beta,\gamma_2}(U)(s)ds} f^i(U), \quad i = 1,2 \\ \dfrac{2\alpha}{\int\limits_S \Xi_{\beta,\gamma_2}(U)(s)ds} f^i(U), \quad i = 3,\ldots,m+2 \end{cases}, \tag{4.13}$$

while every equation in the system is considered separately. To show the next fact, we use the interior $\mathbb{H}^2$-regularity theorem (see [19], Section 6.3.1, Theorem 1), which, in our present notation, reads:

Theorem 4.1 *Assume that $u \in \mathbb{H}^1(S)$ is a weak solution of the elliptic PDE*

$$\Delta u = f \quad in\ S,$$

and $f \in \mathbb{L}^2(S)$. Then $u \in \mathbb{H}^2_{loc}(S)$.

Proof: See [19].

■

That is, there exists the weak solution $U_\beta \in \mathbb{H}^1(S,E)$ of (4.13), s.t.

$$U^i_\beta \in \mathbb{H}^2_{loc}(S), \quad i = 1,\ldots,m+2.$$

Our intention next is to iterate the argument above, thereby deducing our weak solution lies in various higher Sobolev spaces, provided the righthand side lies in sufficiently good spaces. To show this, we use the higher interior regularity theorem ([19], Section 6.3.1, Theorem 2) and general Sobolev inequalities theorem ([19], Section 5.6.3, Theorem 6), which, in our present notation, read respectively in a following manner:

Theorem 4.2 *Assume that $u \in \mathbb{H}^1(S)$ is a weak solution of the elliptic PDE*

$$\Delta u = f \quad in\ S,$$

and $f \in \mathbb{H}^n(S)$. Then $u \in \mathbb{H}^{n+2}_{loc}(S)$.

Proof: See [19].

■

Theorem 4.3 *Let S be a bounded open subset of $\mathbb{R}^2$, with C^1 boundary. Assume $u \in \mathbb{H}^n_{loc}(S)$, $n \geq 2$. Then $u \in C^{n-2}(S)$.*

Proof: See [19].

■

The following proposition will serve us as further induction basis.

Proposition 4.2 *Assume that there exists a weak solution U_β of (4.13), s.t. $U^i_\beta \in \mathbb{H}^1(S)$ for $i = 1, \ldots, m+2$. Then U^i_β, $\Xi_{\beta,\gamma_2}(U_\beta)$ and $e^{-|U_\beta(\cdot)-G(\cdot')|^2_{\gamma_2}/\beta}$ for all $s' \in S$ are in $\mathbb{H}^2_{loc}(S) \cap C^1(S)$.*

Proof: Given in Appendix A.

■

The following proposition includes an induction assumption and its proof.

Proposition 4.3 *Assume that there exists a weak solution U_β of (4.13), s.t. $U^i_\beta \in \mathbb{H}^n_{loc}(S) \cap C^{n-1}(S)$ for $i = 1, \ldots, m+2$. Then U^i_β, $\Xi_{\beta,\gamma_2}(U_\beta)$ and $e^{-|U_\beta(\cdot)-G(\cdot')|^2_{\gamma_2}/\beta}$ for all $s' \in S$ are in $\mathbb{H}^{n+1}_{loc}(S) \cap C^n(S)$.*

Proof: Given in Appendix B.

■

We can now repeatedly apply the Proposition 4.3 for $n = 2, 3, \ldots$ to deduce the infinite differentiability of U_β in the interior, i.e. $U_\beta \in C^\infty(S, E)$.

4.3 Some properties of the minimizer of original functional

In this section we shall discuss some local properties of the minimizer $\overline{U}$ of original functional (3.13):

$$\widetilde{F}(U) = \frac{1}{2} \int_S Tr\left(\nabla U(s)^T h \nabla U(s)\right) ds + \alpha \mathcal{D}^2(\mu, U).$$

Actually, we intend to show here, that $\overline{U}$ is locally Lipschitz continuous in S. We shall base our reasoning on Characterization of $W^{1,\infty}$ Theorem ([19], Section 5.8.2(b), Theorem 4), that, in our present notation, read as follows:

Theorem 4.4 *Let S be open and bounded, with ∂S of class C^1. Then $U : S \to E$ is Lipschitz continuous if and only if $U \in W^{1,\infty}(S, E)$.*

Proof: See [19].

■

Let $\overline{U}$ be a minimizer of the functional $\widetilde{F}(U)$. By the compactness of $supp(\mu)$ and the definition of $\mathcal{D}(\mu, U)$ we obtain that $\overline{U}$ is bounded in $\mathbb{L}^\infty(S, E)$. We tend to show now that $D\overline{U}$ is also bounded in $\mathbb{L}^\infty(S, E)$. To conclude this, we go back to section 3.5 and remind the definition of the sequence of functionals

$$\widetilde{F}_\beta(U) = \frac{1}{2}\int\limits_S Tr\left(\nabla U(s)^T h \nabla U(s)\right) ds + \alpha\beta \ln\left[\int\limits_S \frac{ds}{\int\limits_E e^{-|U(s)-z|^2/\beta}\mu(dz)}\right],$$

converging in Γ-sense to the original functional $\widetilde{F}(U)$. By Theorem 3.5, the sequence of its minimizers U_β weakly converges in $\mathbb{H}^1(S, E)$ to the minimizer of original functional $\overline{U}$. That is, DU_β weakly converges in $\mathbb{L}^2(S, E)$ to $D\overline{U}$.

To show the next fact, we use the theorem dealing with the gradient estimates for Poisson's equation ([20], Section 3.4, Theorem 3.9):

Theorem 4.5 *Let $u \in C^2(\Omega)$ satisfy Poisson's equation, $\Delta u = f$, in Ω. Then*

$$\sup_\Omega d_x |Du(x)| \le C(\sup_\Omega |u| + \sup_\Omega d_x^2 |f(x)|),$$

where $d_x = dist(x, \partial\Omega)$.

Proof: See [20].

■

The following theorem is the main goal of this section.

Theorem 4.6 *Let $\overline{U}$ be a minimizer of functional (3.13). Then $\overline{U}$ is locally Lipschitz continuous in S.*

Proof: We remind, that U_β is the weak solution of system of semi-linear elliptic equations (4.13). By Proposition 4.1, Corollary 4.1, (4.12) and Theorem 4.5, the sequence DU_β is in $\mathbb{L}^\infty_{loc}(S, E)$, and moreover, uniformly bounded there. Thus, by duality of $\mathbb{L}^\infty$ to $\mathbb{L}^1$, we state that for every $i = 1, \ldots, m+2$ and for every $S' \subset\subset S$ there exists

$$\|DU^i_\beta\|_{\mathbb{L}^\infty(S')} = \sup_{\phi \in \mathbb{L}^1(S'),\ \|\phi\|_{\mathbb{L}^1}=1} \int\limits_{S'} DU^i_\beta(s)\phi(s)ds \le M.$$

Then, for every $\phi \in \mathbb{L}^1(S')$, s.t. $\|\phi\|_{\mathbb{L}^1} = 1$,

$$\int\limits_{S'} DU^i_\beta(s)\phi(s)ds \le M. \tag{4.14}$$

Weak convergence of DU_β to $D\overline{U}$ implies

$$\lim_{\beta\to 0}\int\limits_{S'} DU_\beta^i(s)\phi(s)ds = \int\limits_{S'} D\overline{U}^i(s)\phi(s)ds. \tag{4.15}$$

From (4.14) and (4.15) we obtain that

$$\int\limits_{S'} D\overline{U}^i(s)\phi(s)ds \le M$$

for every $\phi \in \mathbb{L}^1(S')$, s.t. $\|\phi\|_{\mathbb{L}^1} = 1$, especially for that one, where $\int\limits_{S'} D\overline{U}^i(s)\phi(s)ds$ attains its supremum. Thus, for every $i = 1, \ldots, m+2$ and for every $S' \subset\subset S$

$$\|D\overline{U}^i\|_{\mathbb{L}^\infty(S')} = \sup_{\phi\in\mathbb{L}^1(S'),\ \|\phi\|_{\mathbb{L}^1}=1}\int\limits_{S'} D\overline{U}^i(s)\phi(s)ds \le M.$$

Finally, $D\overline{U} \in \mathbb{L}^\infty_{loc}(S,E)$. Taking also into account that $\overline{U} \in \mathbb{L}^\infty(S,E)$, we conclude that $\overline{U} \in W^{1,\infty}_{loc}(S,E)$. Thus, by Theorem 4.4, $\overline{U}$ is locally Lipschitz continuous in S.

■

Chapter 5

Non-parametric representation

5.1 Variational formulation and Euler-Lagrange equation for the relaxed functional

Let consider a non-parametric representation of the image U. In this case $U(S) := \{U_B, U_{\mathbf{Y}}\}$, where U_B is the projection of U on the pixel space B and $U_{\mathbf{Y}}$ is the projection of U on the feature fiber $\mathbf{Y}$. Here we identify S with B and set U_B as identity map. Thus, the image is given in terms of a graph $U_{\mathbf{Y}} := f : B \to \mathbf{Y}$.

Considering the image U in terms of a graph of function f does not allow existence of edges in form of the surface folding. Moreover, in case of non-parametric formulation we cannot use the majorant area functional $\widetilde{A}(U)$ as defined in (3.4). For such formulation the area functional is given as in (3.2):

$$A(U) = \int_S |\Gamma(U)(s)| ds = \int_B \sqrt{\gamma^2 + \gamma |\nabla f|^2} db.$$

We also separate the scale γ into γ_1 for the area term and γ_2 for the metric. In order to unify the limits $\gamma_1 \to 0$ and $\gamma_1 \to \infty$ we normalize the area term by $\dfrac{1+\sqrt{\gamma_1}}{\sqrt{\gamma_1}}$. Thus, the corresponding functional is

$$\mathbf{F}^{\beta}_{\gamma_1,\gamma_2}(f) = (1+\sqrt{\gamma_1}) \int_B \sqrt{\gamma_1 + |\nabla f|^2} db + \alpha \mathcal{D}^2_{\beta,\gamma_2}(\mu, f).$$

We define now the relaxed metric term $\mathcal{D}_{\beta,\gamma_2}(\mu, f)$. For non-parametric formulation we represent the data $\mu(dz)$ as $\mu_b(dy)db$. Under the above assumption, the parameter dependent metric (3.22) attributes the measure db to the image f, and it is written as

$$\mathcal{D}^2_{\beta,\gamma_2}(\mu, f) = \beta \ln \left[\int_B \Psi^f_{\beta,\gamma_2}(b) db \right], \tag{5.1}$$

where

$$\Psi^f_{\beta,\gamma_2}(b) = \frac{1}{\int\limits_{\mathbf{Y}}\int\limits_{B} e^{-[|f(b)-y|^2+\gamma_2|b-b'|^2]/\beta} db' \mu_{b'}(dy)}. \tag{5.2}$$

In addition, the following analysis is performed for deterministic data, thus $\Psi^f_{\beta,\gamma_2}(b)$ in (5.2) is given as

$$\Psi^f_{\beta,\gamma_2}(b) = \frac{1}{\int\limits_{B} e^{-[|f(b)-g(b')|^2+\gamma_2|b-b'|^2]/\beta} db'}. \tag{5.3}$$

Finally, the corresponding functional is:

$$\mathbf{F}^{\beta}_{\gamma_1,\gamma_2}(f) = (1+\sqrt{\gamma_1})\int\limits_{B}\sqrt{\gamma_1+|\nabla f|^2}db + \alpha\beta\ln\left[\int\limits_{B}\Psi^f_{\beta,\gamma_2}(b)db\right], \tag{5.4}$$

where $\Psi^f_{\beta,\gamma_2}(b)$ is given as in (5.3).

We aim to determine now the condition for the first variation of the functional (5.4) to be 0. This is a necessary condition for f to minimize it. The minimum is the function f which solves the boundary value problem

$$\begin{aligned} -\mathrm{div}\left(\frac{(1+\sqrt{\gamma_1})\nabla f}{\sqrt{\gamma_1+|\nabla f|^2}}\right) &+ \frac{2\alpha\Psi^f_{\beta,\gamma_2}(b)}{\int\limits_{B}\Psi^f_{\beta,\gamma_2}(b)db} f \\ &- \frac{2\alpha\left(\Psi^f_{\beta,\gamma_2}(b)\right)^2}{\int\limits_{B}\Psi^f_{\beta,\gamma_2}(b)db}\int\limits_{B} g(b')e^{-[|f(b)-g(b')|^2+\gamma_2|b-b'|^2]/\beta}db' = 0 \end{aligned} \tag{5.5}$$

with Neumann boundary condition

$$\frac{\partial f}{\partial \vec{n}} = 0 \quad \text{on } \partial B. \tag{5.6}$$

Note that the function $G(b) = \dfrac{(1+\sqrt{\gamma_1})}{\sqrt{\gamma_1+|\nabla f|^2}}$ in the first term of the left-hand part of (5.5) may be considered as edge indicator (penalty) function.

5.2 Numerical results for one-dimensional problem

We use the Euler-Lagrange equation (5.5) to obtain the following gradient descent flow for an initially chosen function $f(b;t=0)$:

$$\partial_t f = -\frac{\delta\mathbf{F}^{\beta}_{\gamma_1,\gamma_2}(f)}{\delta f} = \mathrm{div}\left(\frac{(1+\sqrt{\gamma_1})\nabla f}{\sqrt{\gamma_1+|\nabla f|^2}}\right) - \frac{2\alpha\Psi^f_{\beta,\gamma_2}(b)}{\int\limits_{B}\Psi^f_{\beta,\gamma_2}(b)db} f$$

$$+ \quad \frac{2\alpha\left(\Psi^f_{\beta,\gamma_2}(b)\right)^2}{\int\limits_B \Psi^f_{\beta,\gamma_2}(b)db}\int\limits_B g(b')e^{-[|f(b)-g(b')|^2+\gamma_2|b-b'|^2]/\beta}db'. \tag{5.7}$$

An initial function $f(b;0)$ (every suitable choice is acceptable) is expected to converge to minimum of the functional (5.4) via the above flow, while keeping on reflecting boundary conditions (5.6).

Consider now the one-dimensional case of (5.7) with boundary condition (5.6) on interval $[a,b]$. Actually, for given $g:\mathbb{R}\to\mathbb{R}$ we attempt to solve the following PDE problem for function $f:\mathbb{R}\to\mathbb{R}$:

$$\begin{aligned}\partial_t f &= \partial_x\left(\frac{(1+\sqrt{\gamma_1})\partial_x f}{\sqrt{\gamma_1+(\partial_x f)^2}}\right) - \frac{2\alpha\Psi^f_{\beta,\gamma_2}(x)}{\int\limits_a^b \Psi^f_{\beta,\gamma_2}(x)dx}f(x)\\ &+ \frac{2\alpha\left(\Psi^f_{\beta,\gamma_2}(x)\right)^2}{\int\limits_a^b \Psi^f_{\beta,\gamma_2}(x)dx}\int\limits_a^b g(x')e^{-[(f(x)-g(x'))^2+\gamma_2(x-x')^2]/\beta}dx',\end{aligned} \tag{5.8}$$

while the boundary conditions are

$$f'(a)=f'(b)=0. \tag{5.9}$$

The following discretization process of (5.8)-(5.9) is fully inspired by [11]. We divide the interval $[a,b]$ on N equal parts by points $x_i=a+ih$, where $h=\dfrac{b-a}{N}$ is a grid size and $0\le i\le N$. Consider also discrete times $t_k=k\tau$, where $k\in\mathbb{N}$ and τ is the time step size. We thus introduce the lattice of coordinates $(a+ih,k\tau)$, on which $f_i^k=f(x_i,t_k)$ and $a_i^k=\dfrac{1+\sqrt{\gamma_1}}{\sqrt{\gamma_1+(\partial_x f(x_i,t_k))^2}}$. Then we discretize $\dfrac{(1+\sqrt{\gamma_1})\partial_x f}{\sqrt{\gamma_1+(\partial_x f)^2}}$ by $a_i^k\partial_x f(x_i,t_{k+1})$ and $\partial_x\left(\dfrac{(1+\sqrt{\gamma_1})\partial_x f}{\sqrt{\gamma_1+(\partial_x f)^2}}\right)$ by

$$\frac{1}{2h^2}\left[(a_{i-1}^k+a_i^k)f_{i-1}^{k+1}-(2a_i^k+a_{i-1}^k+a_{i+1}^k)f_i^{k+1}+(a_i^k+a_{i+1}^k)f_{i+1}^{k+1}\right].$$

We obtain the following semi-implicit scheme in matrix-vector form

$$\frac{f^{k+1}-f^k}{\tau}=A\left(f^k\right)f^{k+1}-\alpha D\left(f^k\right)f^{k+1}+\alpha B\left(f^k\right), \tag{5.10}$$

where

$$\begin{aligned}f_i^0 &= g(x_i) \text{ for } 0\le i\le N,\\ f_0^k &= f_1^k,\ f_{N-1}^k=f_N^k,\end{aligned}$$

$$[A(f^k)]_{ij} = \left\{ \begin{array}{rl} -\dfrac{a_{i-1}^k + 2a_i^k + a_{i+1}^k}{2h^2}, & i = j \\ \dfrac{a_i^k + a_j^k}{2h^2}, & i = j+1,\ j-1 \\ 0, & \text{else} \end{array} \right\} - (N+1) \times (N+1) \text{ matrix},$$

$$[D(f^k)]_{ij} = \left\{ \begin{array}{rl} \dfrac{2\Psi_{\beta,\gamma_2}^{f^k}(a+ih)}{\int\limits_a^b \Psi_{\beta,\gamma_2}^f(b)db}, & i = j \\ 0, & i \neq j \end{array} \right\} - (N+1) \times (N+1) \text{ matrix},$$

$$[B(f^k)]_i = \frac{2\left(\Psi_{\beta,\gamma_2}^{f^k}(a+ih)\right)^2}{\int\limits_a^b \Psi_{\beta,\gamma_2}^f(b)db} \int\limits_a^b g(b')e^{-[|f(a+ih)-g(b')|^2+\gamma_2|a+ih-b'|^2]/\beta}db'$$

– vector of length $N+1$.

Here by $\int\limits_a^b \Psi_{\beta,\gamma_2}^f(b)db$ and $\int\limits_a^b g(b')e^{-[|f(a+ih)-g(b')|^2+\gamma_2|a+ih-b'|^2]/\beta}db'$ we mean the approximations of respective integrals via the trapezoidal method.

Finally, the equation (5.10) can be written as $(N+1) \times (N+1)$ linear system

$$\left[I - \tau\left(A\left(f^k\right) - \alpha D\left(f^k\right)\right)\right] f^{k+1} = f^k + \tau\alpha B\left(f^k\right). \tag{5.11}$$

Note that the matrix $A\left(f^k\right)$ is tridiagonal and positive definite, the matrix $D\left(f^k\right)$ is diagonal, thus the matrix $I - \tau\left(A\left(f^k\right) - \alpha D\left(f^k\right)\right)$ is invertible and the solution f^{k+1} of (5.11) can be written as

$$f^{k+1} = \left[I - \tau\left(A\left(f^k\right) - \alpha D\left(f^k\right)\right)\right]^{-1}\left(f^k + \tau\alpha B\left(f^k\right)\right). \tag{5.12}$$

The most efficient way to solve the tridiagonal linear system (5.12) is the so-called *Thomas algorithm*, a Gaussian elimination algorithm for tridiagonal systems (see [56]). It is absolutely stable for every strictly diagonally dominant system matrix, and consequently, the large time steps may be used.

We end up this section with suitable numerical experiments for one variable functions. In all cases the domain $[0, 6]$ was descretized by 200 points, $\beta = 0.0005$ and $\gamma_2 = 10^{16}$.At first, the Gaussian

$$f(x) = \frac{1}{\sqrt{2\pi}} \exp^{-\frac{(x-3)^2}{2}},$$

replaced at the ends by piecewise constant functions was segmented by various combinations of parameters (see Figure 5.1).

Eventually, the previous function (Gaussian with piecewise constant functions at the ends) was perturbed in the middle by sinus function with big (10π) period. The various forms of its segmentation are introduced in Figure 5.2.

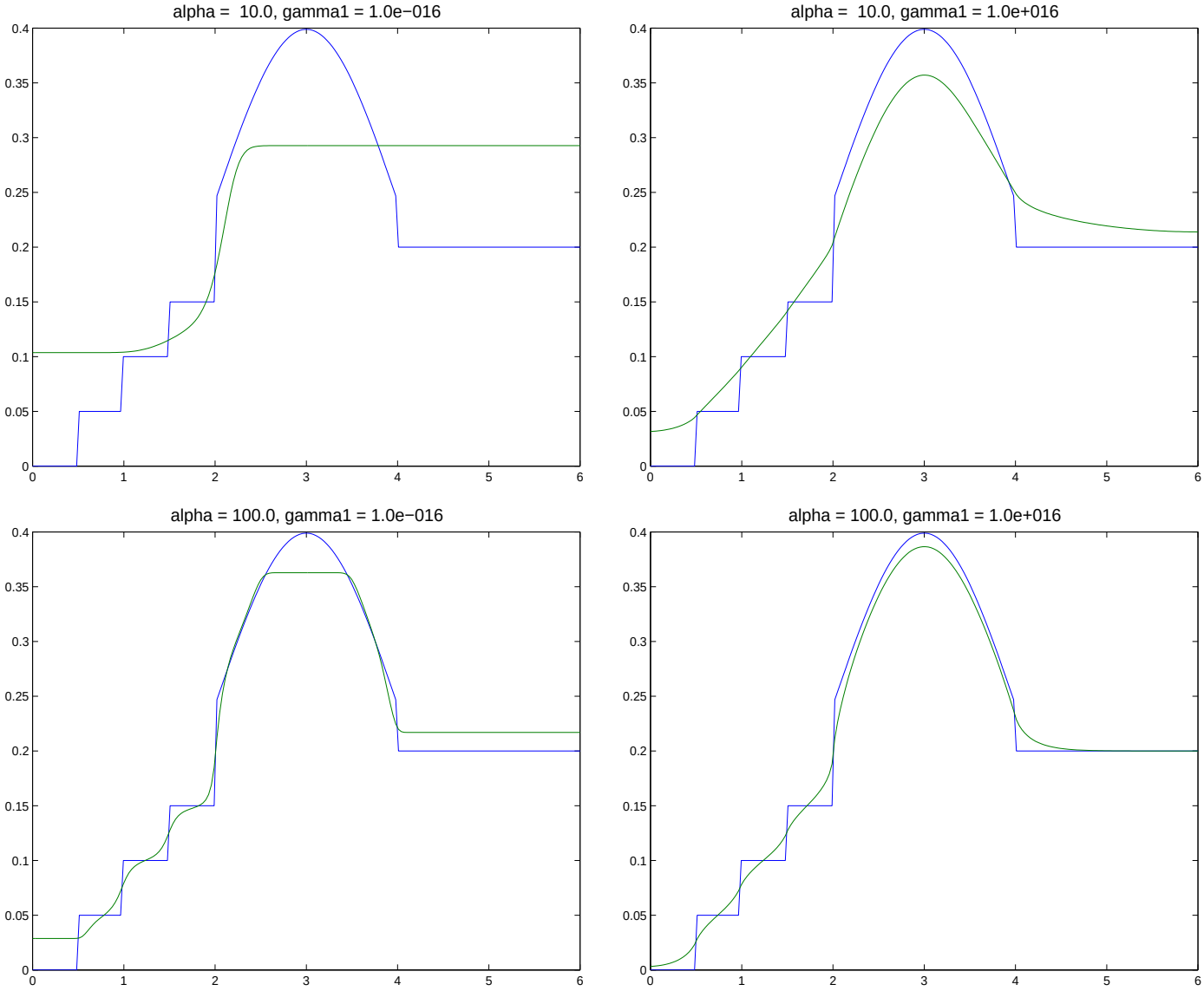


Figure 5.1: Segmentation of modified Gaussian ($\alpha = 10$ in upper row, $\alpha = 100$ in lower row) with TV norm (left column) and H^1 norm (right column).

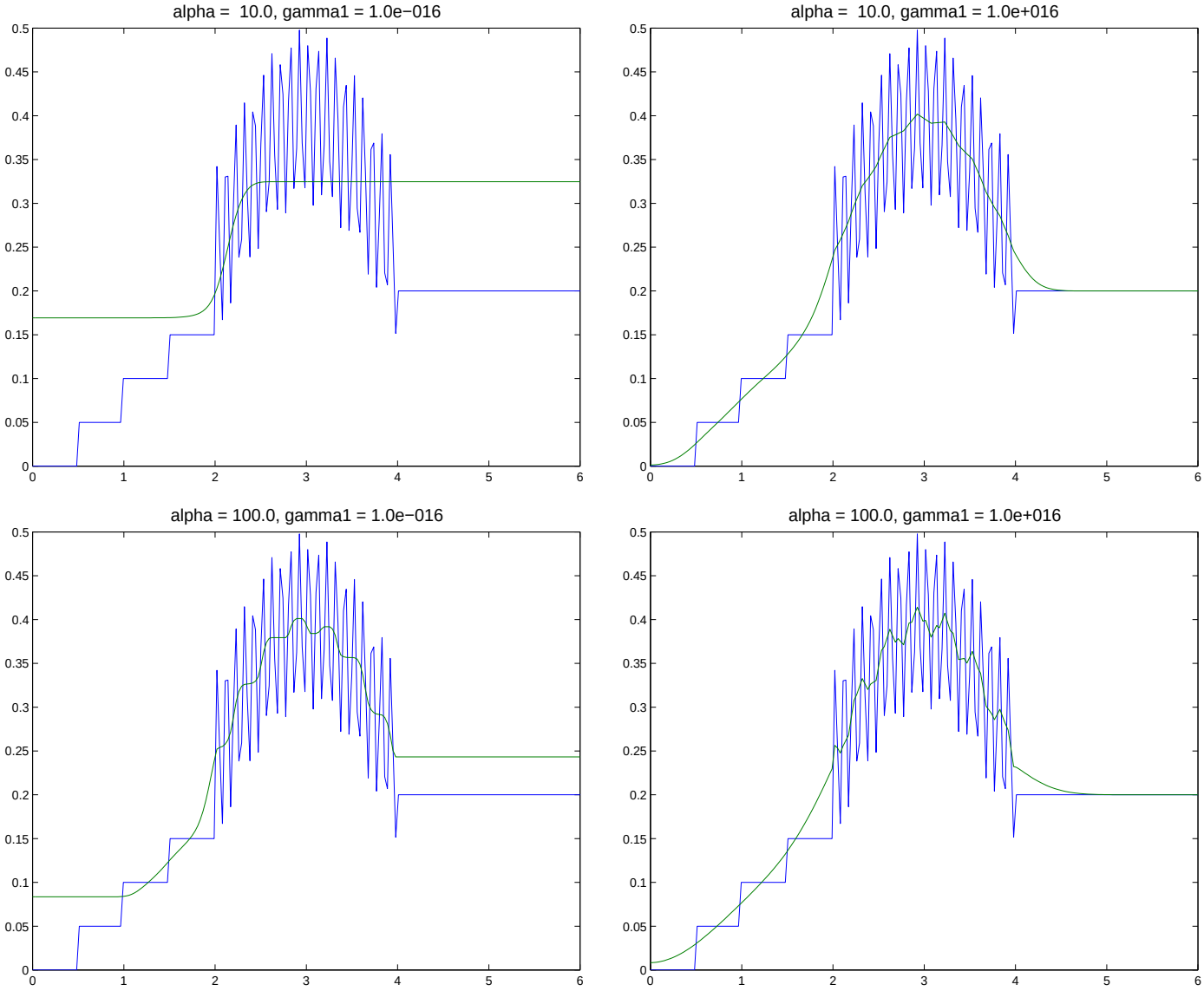


Figure 5.2: Segmentation of perturbed Gaussian ($\alpha = 10$ in upper row, $\alpha = 100$ in lower row) with TV norm (left column) and H^1 norm (right column).

5.3 Numerical implementation of two-dimensional problem

Consider now the $M \times N$ image (M rows and N columns). Let $B \subset \mathbb{R}^2$ be its domain and $g(b),\ b \in B$ denote its graph. We assume that its desired segmentation should be the minimum of relaxed functional (5.4), i.e.

$$\mathbf{F}^{\beta}_{\gamma_1,\gamma_2}(f) = (1+\sqrt{\gamma_1}) \int\limits_B \sqrt{\gamma_1 + |\nabla f|^2} db + \alpha\beta \ln \left[\int\limits_B \Psi^f_{\beta,\gamma_2}(b) db \right], \tag{5.13}$$

where $\Psi^f_{\beta,\gamma_2}(b)$ is given as in (5.3). We are looking now for a weak solution of the following Euler-Lagrange equation

$$\begin{aligned} -\text{div}\left(\frac{(1+\sqrt{\gamma_1})\nabla f}{\sqrt{\gamma_1 + |\nabla f|^2}} \right) \quad &+ \quad \frac{2\alpha \Psi^f_{\beta,\gamma_2}(b)}{\int\limits_B \Psi^f_{\beta,\gamma_2}(b) db} f(b) \\ &- \quad \frac{2\alpha \left(\Psi^f_{\beta,\gamma_2}(b) \right)^2}{\int\limits_B \Psi^f_{\beta,\gamma_2}(b) db} \int\limits_B g(b') e^{-[|f(b)-g(b')|^2 + \gamma_2 |b-b'|^2]/\beta} db' = 0. \end{aligned} \tag{5.14}$$

with natural boundary conditions

$$\left. \frac{\partial f}{\partial \vec{n}} \right|_{\partial B} = 0. \tag{5.15}$$

There are some existing numerical methods for solving the above nonlinear type PDEs, for instance, time marching [48], lagged diffusivity fixed point (FP) schemes [55] and primal-dual methods [13]. Due to the robustness and simplicity of implementation of the fixed point algorithm, we apply it to solve (5.14)-(5.15) in this work. The idea of the FP method is to first linearize the nonlinear PDE (5.14) by lagging the diffusive coefficient $\frac{(1+\sqrt{\gamma_1})}{\sqrt{\gamma_1 + |\nabla f|^2}}$ and the relevant parts of right -hand side of (5.14) (like $\Psi^f_{\beta,\gamma_2}(b)$ and $e^{-[|f(b)-g(b')|^2+\gamma_2|b-b'|^2]/\beta}$) by one iteration, and then apply the fixed point method to solve linear problem for f. More precisely, the FP iterative method is described as follows:

- *Assume we have* f^n.
- *Solve for* f^{n+1}

$$\begin{aligned} -\text{div}\left(\frac{(1+\sqrt{\gamma_1})\nabla f^{n+1}}{\sqrt{\gamma_1 + |\nabla f^n|^2}} \right) \quad &+ \quad \frac{2\alpha \Psi^{f^n}_{\beta,\gamma_2}(b)}{\int\limits_B \Psi^{f^n}_{\beta,\gamma_2}(b) db} f^{n+1}(b) \\ &- \quad \frac{2\alpha \left(\Psi^{f^n}_{\beta,\gamma_2}(b) \right)^2}{\int\limits_B \Psi^{f^n}_{\beta,\gamma_2}(b) db} \int\limits_B g(b') e^{-[|f^n(b)-g(b')|^2 + \gamma_2 |b-b'|^2]/\beta} db' = 0. \end{aligned}$$

The following discretization process of (5.14)-(5.15) is fully inspired by [11]. We introduce the lattice of coordinates (ih_y, jh_x), where

$$h_y = \frac{1}{M-1},\ h_x = \frac{1}{N-1},\ 0 \le i \le M-1,\ 0 \le j \le N-1.$$

We denote by $f^n_{i,j}$ the $n-$th FP iteration of f at point (ih_y, jh_x) and by $a^n_{i,j}$ an approximation of $\dfrac{1+\sqrt{\gamma_1}}{\sqrt{\gamma_1+|\nabla f^n(ih_y,jh_x)|^2}}$. Then we discretize $\dfrac{(1+\sqrt{\gamma_1})\partial_x f}{\sqrt{\gamma_1+|\nabla f|^2}}$ by $a^n_{i,j}\partial_x f^{n+1}(ih_y, jh_x)$ and $\partial_x\left(\dfrac{(1+\sqrt{\gamma_1})\partial_x f}{\sqrt{\gamma_1+|\nabla f|^2}}\right)$ by

$$\frac{1}{2h_x^2}\left[(a^n_{i,j-1}+a^n_{i,j})f^{n+1}_{i,j-1} - (a^n_{i,j-1}+2a^n_{i,j}+a^n_{i,j+1})f^{n+1}_{i,j} + (a^n_{i,j}+a^n_{i,j+1})f^{n+1}_{i,j+1}\right].$$

In the same way we discretize $\dfrac{(1+\sqrt{\gamma_1})\partial_y f}{\sqrt{\gamma_1+|\nabla f|^2}}$ by $a^n_{i,j}\partial_y f^{n+1}(ih_y, jh_x)$ and $\partial_y\left(\dfrac{(1+\sqrt{\gamma_1})\partial_y f}{\sqrt{\gamma_1+|\nabla f|^2}}\right)$ by

$$\frac{1}{2h_y^2}\left[(a^n_{i-1,j}+a^n_{i,j})f^{n+1}_{i-1,j} - (a^n_{i-1,j}+2a^n_{i,j}+a^n_{i+1,j})f^{n+1}_{i,j} + (a^n_{i,j}+a^n_{i+1,j})f^{n+1}_{i+1,j}\right].$$

We also denote by $b^n_{i,j}$ an approximation of $\dfrac{\Psi^{f^n}_{\beta,\gamma_2}(ih_y,jh_x)}{\int\limits_B \Psi^{f^n}_{\beta,\gamma_2}(b)db}$ and by $c^n_{i,j}$ an approximation of

$$\frac{\left(\Psi^{f^n}_{\beta,\gamma_2}(ih_y,jh_x)\right)^2}{\int\limits_B \Psi^{f^n}_{\beta,\gamma_2}(b)db}\int\limits_B g(b')e^{-[|f^n(ih_y,jh_x)-g(b')|^2+\gamma_2|(ih_y,jh_x)-b'|^2]/\beta}db'.$$

The last approximations imply computation of definite integrals. Due to its integrands being the piecewise constant functions, its computation is the summation of appropriate function values on lattice grid multiplied by the value of grid element. For example, for $\int\limits_B \Psi^f_{\beta,\gamma_2}(b)db$ we have

$$\int\limits_B \Psi^f_{\beta,\gamma_2}(b)db = \frac{1}{(M-1)\cdot(N-1)}\sum_{i=0}^{M-1}\sum_{j=0}^{N-1}\Psi^f_{\beta,\gamma_2}(ih_y,jh_x).$$

Finally, we obtain the following semi-implicit scheme

$$\begin{aligned}
&- \frac{1}{2h_y^2}(a^n_{i-1,j}+a^n_{i,j})f^{n+1}_{i-1,j} - \frac{1}{2h_x^2}(a^n_{i,j-1}+a^n_{i,j})f^{n+1}_{i,j-1}\\
&+ \left[\frac{1}{2h_y^2}(a^n_{i-1,j}+2a^n_{i,j}+a^n_{i+1,j}) + \frac{1}{2h_x^2}(a^n_{i,j-1}+2a^n_{i,j}+a^n_{i,j+1}) + 2\alpha\cdot b^n_{i,j}\right]f^{n+1}_{i,j}\\
&- \frac{1}{2h_x^2}(a^n_{i,j}+a^n_{i,j+1})f^{n+1}_{i,j+1} - \frac{1}{2h_y^2}(a^n_{i,j}+a^n_{i+1,j})f^{n+1}_{i+1,j} = 2\alpha\cdot c^n_{i,j},
\end{aligned}$$

where

$$\begin{aligned} f^0_{i,j} &= g(ih_y, jh_x) \quad \text{for } 0 \le i \le M-1,\ 0 \le j \le N-1, \\ f^n_{0,j} &= f^n_{1,j},\ f^n_{M-1,j} = f^n_{M-2,j} \quad \text{for } 0 \le j \le N-1, \\ f^n_{i,0} &= f^n_{i,1},\ f^n_{i,N-1} = f^n_{i,N-2} \quad \text{for } 0 \le i \le M-1. \end{aligned}$$

Now the discrete problem can be written in the matrix-vector form as $MN \times MN$ linear system

$$A\left(f^n\right) f^{n+1} = b\left(f^n\right), \tag{5.16}$$

where $A\left(f^n\right)$ is block tridiagonal, symmetric and positive definite(SPD).

In order to solve (5.16), one may apply a variety of iterative methods, like preconditioned conjugate gradients (PCG) methods (see [22]) and multi-grid methods (see [38]). In this work we don't deal with such kind of problem, thus we use here the standard MATLAB function to solve the above linear system for each iteration. The final output, which is the grey-level image, is the desired segmentation. In order to present the results in a pleasant way, the segmentation contour should be obtained. To achieve this, we threshold the accepted segmentation image by **graythresh** MATLAB function using Otsu's method [43], which chooses the threshold to minimize the intra-class variance of the black and white pixels, and covert it to the binary image. Finally, Canny edge detector [8] is applied to the binary image, and achieved segmentation contour is applied to the original image g.

5.4 Segmentation of grey-level images: choice of parameters, limiting cases and numerical results

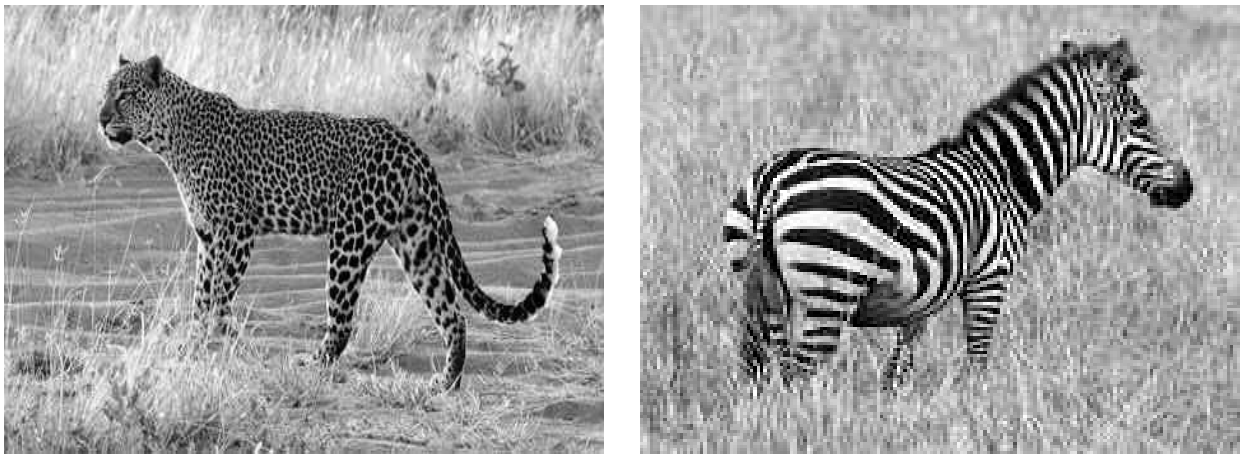


Figure 5.3: Real-life test images of a leopard and a zebra

In this section we are going to point out the influence of parameters $\beta, \alpha, \gamma_1, \gamma_2$ on the segmentations results. To demonstrate the performance of the proposed method, two natural images were used (see Fig. 5.3) in our numerical experiments. We remind that β is the convergence parameter and the minimizer of the original (the limit) functional is achieved when $\beta \to 0$. We also note that the limit $\beta \to 0$ of (5.13) yields

$$\mathbf{F}^0_{\gamma_1,\gamma_2}(f) = (1+\sqrt{\gamma_1}) \int\limits_B \sqrt{\gamma_1 + |\nabla f|^2} db + \alpha \sup_{b\in B} \inf_{b'\in B} \left[|f(b) - g(b')|^2 + \gamma_2 |b-b'|^2 \right],$$

which is the original functional for deterministic images. Thus, for our calculations, β is taken sufficiently small. We show the role of parameters α and γ_1 on the examples of leopard and zebra

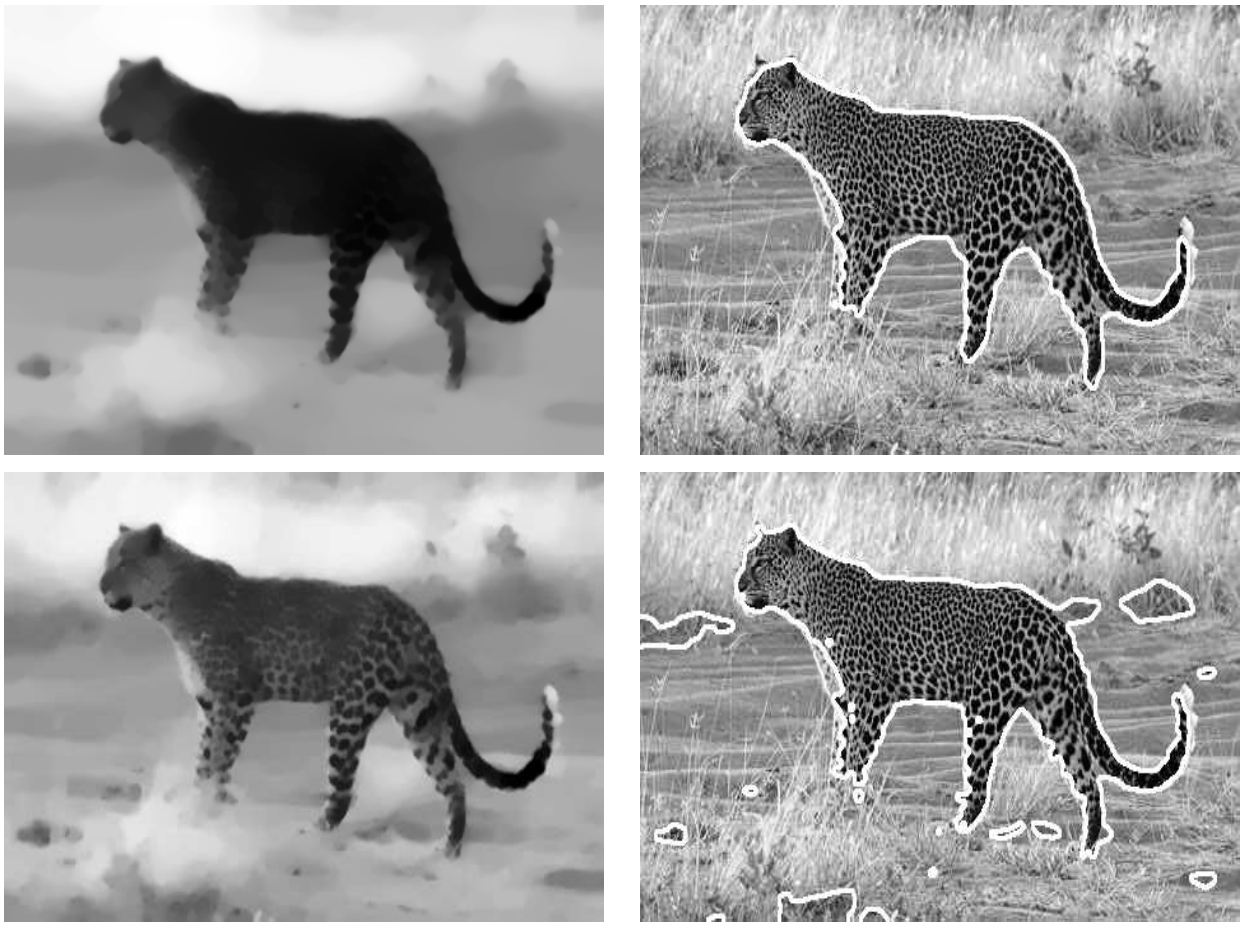


Figure 5.4: Leopard segmentation outputs and outlined original images: $\beta = 0.1,\ \gamma_1 = 0,\ \gamma_2 = \infty$; $\alpha = 185$ in the upper row and $\alpha = 500$ in the lower row.

images respectively, tested in some previous works, especially in [50]. The parameter α, as in case of Mumford-Shah functional, measures the trade off between a good fit of the solution f to data g and the regularity of the solution f. The result obtained for the leopard image (Fig. 5.4) in case

$\alpha = 185$ is comparable to that obtained by Sagiv, Sochen and Zeevi (see [50]).



Figure 5.5: Zebra segmentation outputs: $\alpha = 330,\ \beta = 0.1,\ \gamma_2 = \infty$; $\gamma_1 = 0$ in the upper row on the left, $\gamma_1 = 10$ in the upper row on the right, $\gamma_1 = \infty$ in the lower row on the left and outlined original image for the last case.

We define now two limiting cases for the functional (5.13): $\mathbf{F}^{\beta}_{0,\gamma_2}(f)$ and $\mathbf{F}^{\beta}_{\infty,\gamma_2}(f)$, when $\gamma_1 \to 0$ and $\gamma_1 \to \infty$, respectively. As γ_1 tends to 0, one can obtain

$$\mathbf{F}^{\beta}_{0,\gamma_2}(f) = TV(f) + \alpha\beta \ln \left[\int_B \Psi^{f}_{\beta,\gamma_2}(b) db \right],$$

where

$$TV(f) \equiv \int_B |\nabla f| db$$

is the *Total Variation* (TV) norm, originally introduced by Rudin et. al. in [48]. On the other hand, $\gamma_1 \to \infty$ yields

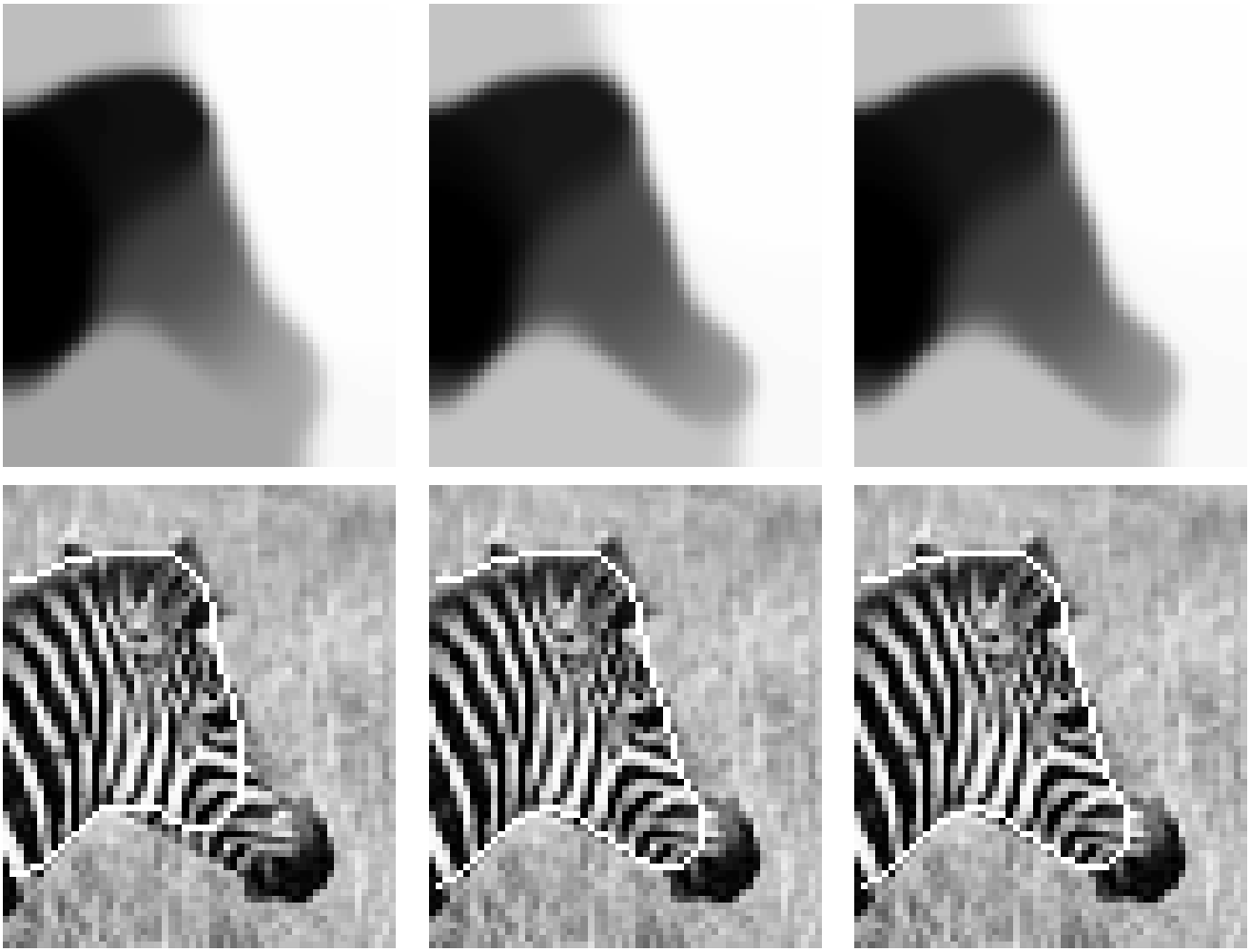


Figure 5.6: Zebra head segmentation outputs (upper row) and their outlined original images (lower row): $\alpha = 5000,\ \beta = 10^{-6},\ \gamma_1 = 0$; $\gamma_2 = 0$ in the left column, $\gamma_2 = 10^4$ in the middle column and $\gamma_2 = 10^8$ in the right column.

$$\mathbf{F}^{\beta}_{\infty,\gamma_2}(f) \approx H^1(f) + \alpha\beta \ln \left[\int\limits_B \Psi^f_{\beta,\gamma_2}(b) db \right]$$

up to the constant, where

$$H^1(f) \equiv \int\limits_B |\nabla f|^2 db$$

is Sobolev space norm (see [58]). We remark that the TV norm allows discontinuities in f, thus making it superior to the H^1 regularization in cases where f can have sharp edges. As is shown above, the parameter γ_1 being defined as representing the scale difference between pixel domain and feature fiber, also determines the kind of regularization for function f, defining the various regularization norms. The results for various cases of parameter γ_1 may be seen in Fig. 5.5.

The role of parameter γ_2, which similarly to parameter γ_1 being defined as representing the scale difference between pixel domain and feature fiber, is completely different. Here the case of interest is the limit $\gamma_2 \to \infty$. It is easily verified that, as $\gamma_2 \to \infty$,

$$\lim_{\gamma_2 \to \infty} \Psi^f_{\beta,\gamma_2}(b) = C\frac{\gamma_2}{\beta} \cdot e^{|f(b)-g(b)|^2/\beta},$$

where C stands for a generic constant. Thus, the limit $\gamma_2 \to \infty$ yields

$$\begin{aligned} \mathbf{F}^{\beta}_{\gamma_1,\infty}(f) = \lim_{\gamma_2 \to \infty} \mathbf{F}^{\beta}_{\gamma_1,\gamma_2}(f) \quad &\approx \quad (1+\sqrt{\gamma_1}) \int\limits_B \sqrt{\gamma_1 + |\nabla f|^2} db \\ &+ \quad \alpha\beta \ln \left[\int\limits_B e^{|f(b)-g(b)|^2/\beta} db \right] - \alpha\beta \ln \beta \end{aligned} \tag{5.17}$$

up to the constant multiplied by β. Finally, we note that the limit $\beta \to 0$ of (5.17) yields

$$\mathbf{F}^{0}_{\gamma_1,\infty}(f) = \lim_{\beta \to 0} \mathbf{F}^{\beta}_{\gamma_1,\infty}(f) = (1+\sqrt{\gamma_1}) \int\limits_B \sqrt{\gamma_1 + |\nabla f|^2} db + \alpha \sup_{b \in B} |f(b) - g(b)|^2.$$

Intuitively speaking, the parameter γ_2 determines the local neighborhood of the pixel b, which gives the valuable contribution to the final segmentation image. On Fig. 5.6 one may see, that on the left upper image ($\gamma_2 = 0$), the region under zebra's head is darker, then on the two other segmentation images for bigger values of γ_2. The explanation is that, the smaller γ_2 is, the larger neighborhood of the given pixel influences on its grey level in the output image. Concerning the above region, it was influenced by the zebra's neck and head (which are mostly dark regions) in case $\gamma_2 = 0$ more than in the two other cases.



Figure 5.7: Zebra segmentation output and outlined original image: $\alpha = 30000,\ \beta = 10^{-6},\ \gamma_1 = 0,\ \gamma_2 = 75000$

Finally, we present the ideal segmentation for the full zebra image (see Fig. 5.7), comparable to that obtained by Sagiv, Sochen and Zeevi (see [50]). It is the best result we got for this image. We

cannot say that it is the absolute best result, since the problem involves four different parameters, and each parameter may have a substantial impact on the final result. However, the point we would like to stress here is the usefulness of giving a finite value to the parameter γ_2.

5.5 Ultrasound image segmentation

Ultrasound imaging is a common, relatively inexpensive and safe modality. It is widely used in medical diagnostics. It allows high acquisition rates and provides images in real-time, but the images are corrupted by the high level of speckle noise. Although the imaging capabilities of modern

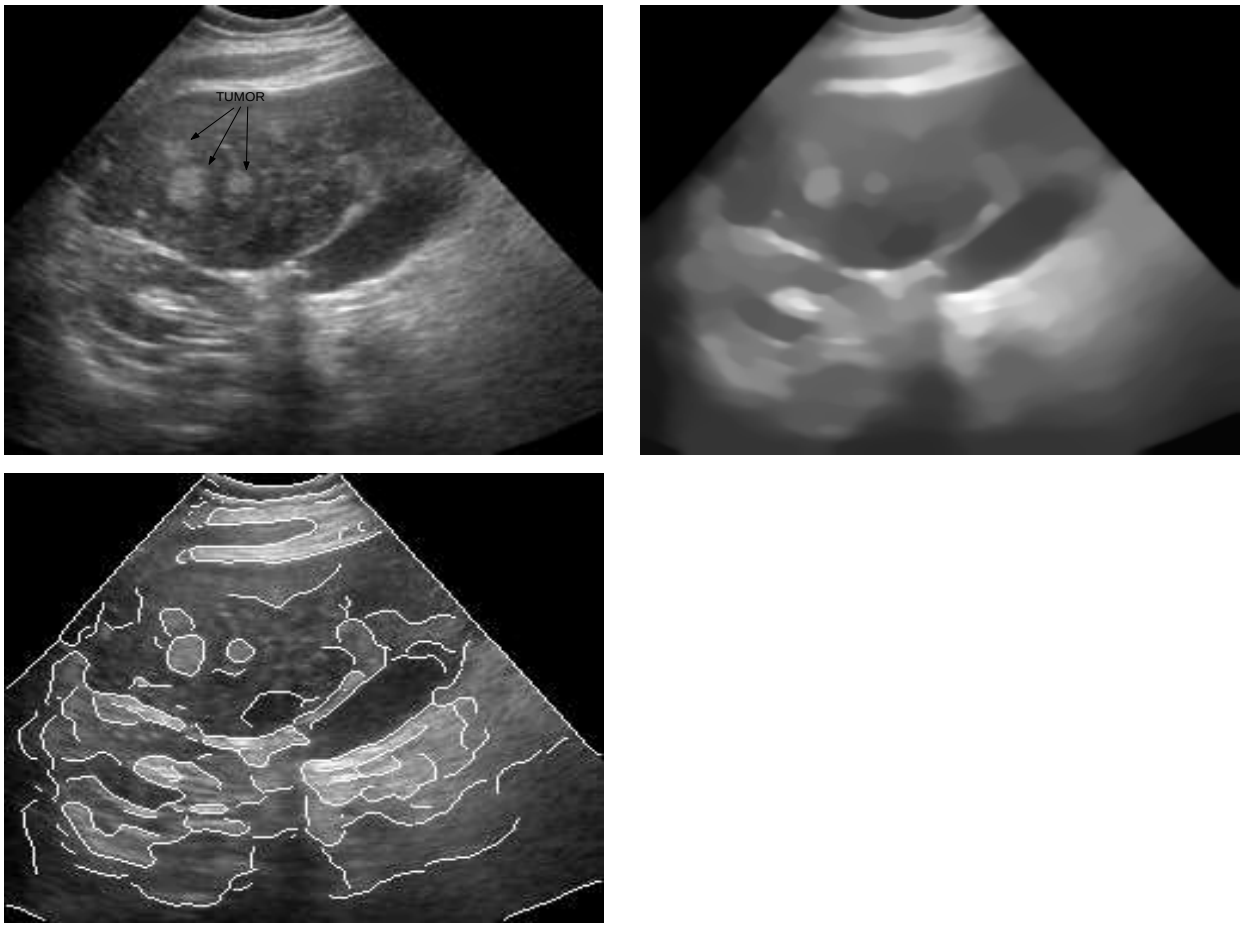


Figure 5.8: Liver tumor segmentation: $\alpha = 650,\ \beta = 0.1,\ \gamma_1 = 0,\ \gamma_2 = \infty$; original image in the upper row on the left, segmentation output in the upper row on the right and outlined original image in the lower row.

ultrasound machines are increasing, noise remains a problem when trying to devise image processing techniques. In fact, noise makes it difficult to accurately identify edges, since in some regions the noise produces artificial edges, while in other regions there are no echoes present and the edges seem

ambiguous. In such low-quality images (which are very common in ultrasound imaging), generic algorithms do not identify the border accurately.

Several algorithms have been reported, which could help identify edges in ultrasound images [3], [12], [17], [25]. Studies that are focused on the processing of cardiac ultrasound images [1], [16] are also of value, since many of the methods are similar. In addition, a modified version [45] for identifying the lumen-thrombus border in case of aortic aneurysm was developed. Some of these studies are based on active contours [26] or "deformable contour" [34], some on the "snakes" algorithm [27], that defines an objective function, also called "energy" function, which is minimized to obtain the estimated border.

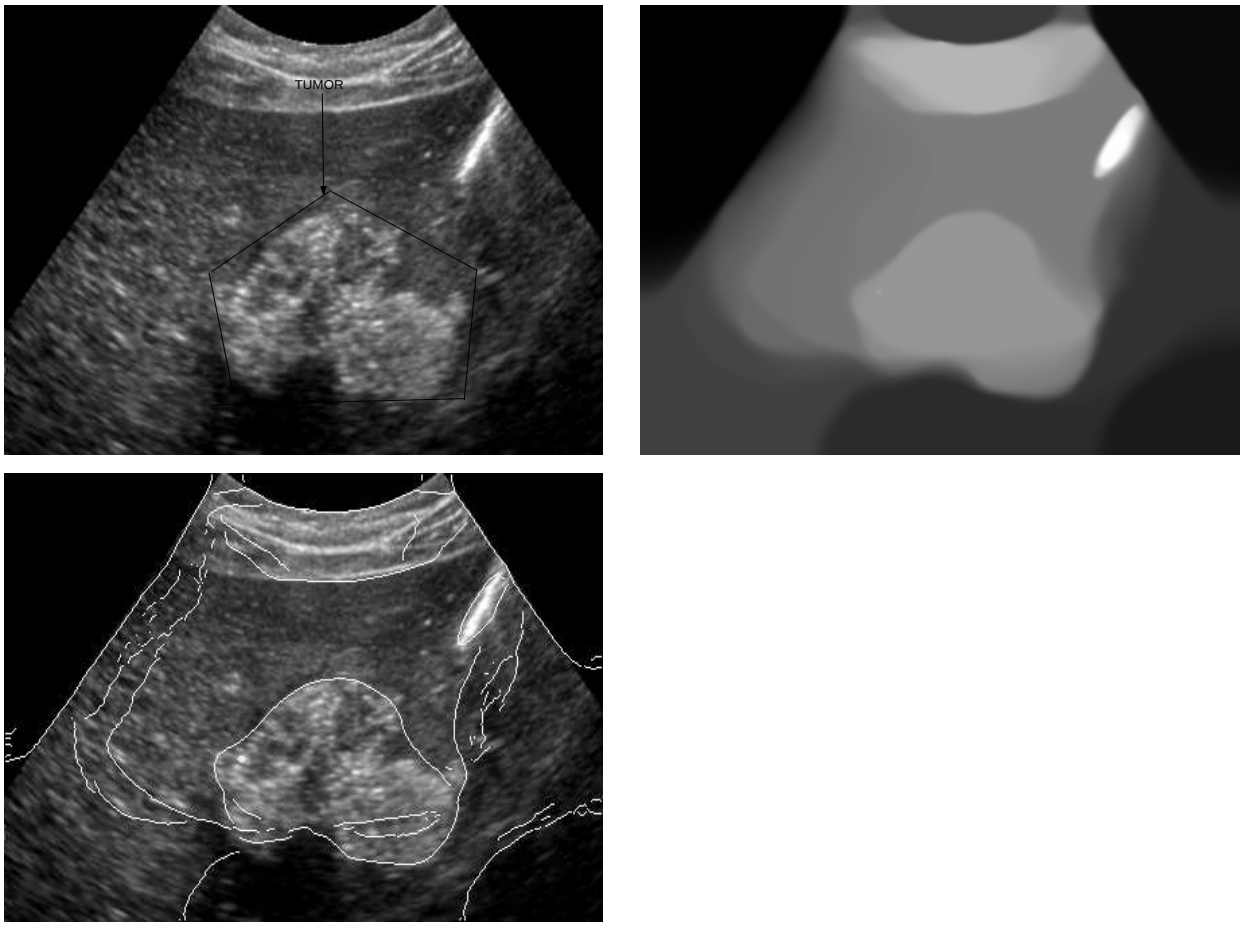


Figure 5.9: Liver tumor segmentation: $\alpha = 50,\ \beta = 0.1,\ \gamma_1 = 0,\ \gamma_2 = \infty$; original image in the upper row on the left, segmentation output in the upper row on the right and outlined original image in the lower row.

We attempt here to apply our objective function (i.e. relaxed functional) to the ultrasonic images of liver containing some kind of tumor. Ultrasound imaging of the liver is a standard part of a checkup by the specialist in internal medicine. These images are convenient to work with due to their high

availability and use in the medical community. The liver is a fairly homogenous organ, in the sense that any given cross-section tends to be fairly uniform. This makes identification of abnormalities somewhat easier. On the other hand, the main difficulty in tumor identification is the fact that the tumor on hepatic ultrasonic images may be presented at different grey levels with respect to other liver tissues. It means that it may be of brighter (see Fig. 5.8) or darker (see Fig. 5.10) grey levels then the healthy tissue of liver, or to include wide variety of existing grey levels (see Fig. 5.9).

The segmentation algorithm is exactly as described in Section 5.3, excluding the thresholding of accepted segmentation image by **graythresh** MATLAB function: Canny edge detector [8] is applied directly to the algorithm output image. Figures 5.8-5.10 present the examples of segmented tumor in hepatic ultrasonic images. In all the cases we used TV regularization norm ($\gamma_1 = 0$) in order to achieve sharper edges and localized metric term ($\gamma_2 = \inf$) in order to reduce the computation time.

On the Fig. 5.10 in addition to tumor, one may see also the segmented kidney and gallbladder.

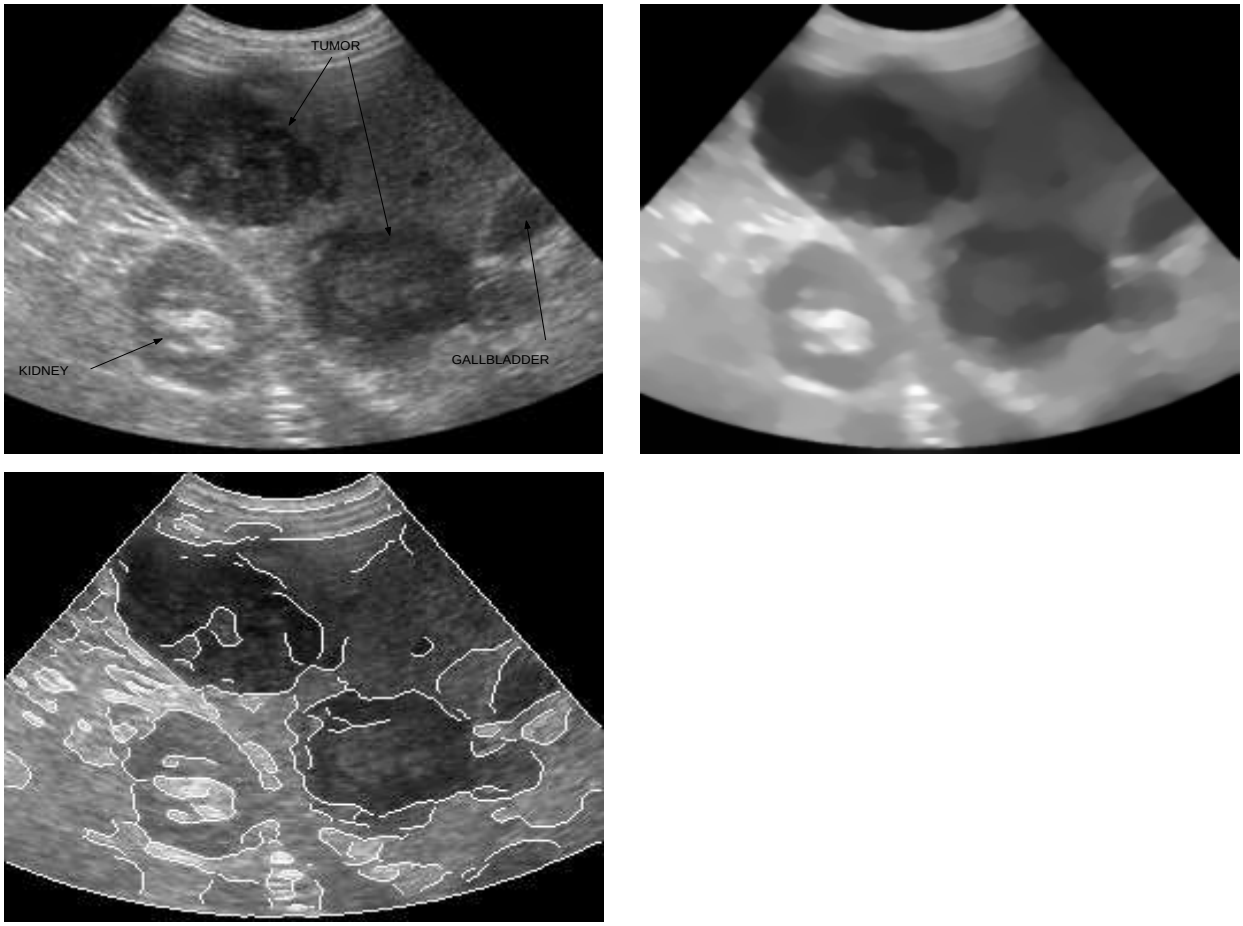


Figure 5.10: Liver tumor segmentation: $\alpha = 900,\ \beta = 0.1,\ \gamma_1 = 0,\ \gamma_2 = \infty$; original image in the upper row on the left, segmentation output in the upper row on the right and outlined original image in the lower row.

Chapter 6

Discussion and conclusions

The importance of the functional, proposed in this study, as an alternative to the widely analyzed and implemented functional of Mumford and Shah, is due to the fact that it is independent of parameterization. It is a geometric functional given in terms of the geometry of imbedded surfaces representing the data and image in the feature space. By combining the Γ-convergence technique with minimal surface theory our approach yielded a global generalization of the Mumford-Shah segmentation functional. Likewise one may consider the implementation of this geometric approach in other image processing and computer vision tasks. In fact, we have illustrated it also in the context of inpainting.

With reference to segmentation of images, it should be pointed out that there exist a variety of segmentation methods, each being suitable for a specific type of segmentation problem. However, there does not exist a universal algorithm that is optimal for the wide spectrum of images and criteria. The present study is a step towards the development of a general framework that can deal with segmentation problems in the context of multi-channel images.

The main novelty of this study is the replacement of the metric term of Mumford-Shah functional by a completely new type of metric based on Hausdorff distance function. This may be useful in cases of defocusing and of mapping problems, where the relevant errors imply that each pixel point is influenced by a domain rather by a single point of a data. Eventually, the proposed change allows us to replace the deterministic data by a random one and include also the case of missing data (inpainting).

Since the new metric term, and thus the functional, suffered from lack of regularity, we utilized an approach adopted from mathematical theory of approximation of functionals via Γ-convergence to overcome this deficiency. The idea is to approximate the functional by a series of different, parameter-dependent functionals, that are expected to be more regular. However, we should point out that the developed relaxed functional demands extensive computational effort to obtain its minimum.

This is the main drawback of our algorithm. An optional solution to this problem is to apply a multi-resolution analysis [7], by performing the relative computations on higher levels of a Gaussian pyramid and thereby reduce significantly the amount of required computations.

So far we have not touched upon the numerical solution problem of data with parametric representation - the case, whose theoretical developments were introduced in Chapter 4. The parametric representation refers to the case of partially free boundary problem in minimal surface theory. The numerical solution may be obtained by finite element method (see, for example, [57]), where the original variational formulation (and not the PDE as in case of non-parametric representation) is used as a point of departure. This yields, however, a relatively complicated variational formulation (due to the relaxed metric term) of our problem in comparison with one treated in [57]. Obtaining the numerical solution for non-parametric case may be the separate theme for future research.

Considering the application of our functional in the context of the inpainting problem., the concept behind our contribution is a modification of the developed metric term.

Appendix A

Proof of Proposition 4.2

We intend to show at first that $f^i(U_\beta) \in \mathbb{H}^1(S),\ i = 1, \ldots, m+2$. Taking into account that

$$D\Xi_{\beta,\gamma_2}(U_\beta)(s) = \frac{2}{\beta}\left(\Xi_{\beta,\gamma_2}(U_\beta)(s)\right)^2 \int\limits_S e^{-|U_\beta(s)-G(s')|^2_{\gamma_2}/\beta} \sum_{k=1}^{m+2} \gamma_2^k \left(U_\beta^k(s) - G^k(s')\right) DU_\beta^k(s)ds',$$

where

$$\gamma_2^k = \begin{cases} \gamma_2, & k = 1,2 \\ 1, & k = 3, \ldots, m+2 \end{cases},$$

from the definition of $f^i(U_\beta)(s)$ we have

$$\begin{aligned} Df^i(U_\beta)(s) &= \frac{4}{\beta}\left(\Xi_{\beta,\gamma_2}(U_\beta)(s)\right)^3 \int\limits_S e^{-|U_\beta(s)-G(s')|^2_{\gamma_2}/\beta} \sum_{k=1}^{m+2} \gamma_2^k \left(U_\beta^k(s) - G^k(s')\right) DU_\beta^k(s)ds' \\ &\cdot \int\limits_S e^{-|U_\beta(s)-G(s')|^2_{\gamma_2}/\beta} \left(U_\beta^i(s) - G^i(s')\right) ds' - \frac{2}{\beta}\left(\Xi_{\beta,\gamma_2}(U_\beta)(s)\right)^2 \\ &\cdot \int\limits_S e^{-|U_\beta(s)-G(s')|^2_{\gamma_2}/\beta} \left(U_\beta^i(s) - G^i(s')\right) \sum_{k=1}^{m+2} \gamma_2^k \left(U_\beta^k(s) - G^k(s')\right) DU_\beta^k(s)ds' \\ &+ \left(\Xi_{\beta,\gamma_2}(U_\beta)(s)\right)^2 \int\limits_S e^{-|U_\beta(s)-G(s')|^2_{\gamma_2}/\beta} DU_\beta^i(s)ds'. \end{aligned}$$

Using the estimate (4.11) and taking into account that $e^{-|U(s)-G(s')|^2_{\gamma_2}/\beta} \le 1$ and $|U_\beta^k(s) - G^k(s')| \le 2R$, we obtain

$$\begin{aligned} \left|Df^i(U_\beta)(s)\right| &\le \frac{4e^{12(2\gamma_2+m)^2R^2/\beta}}{\beta\pi^3} \int\limits_S \sum_{k=1}^{m+2} \gamma_2^k \left|U_\beta^k(s) - G^k(s')\right| \left|DU_\beta^k(s)\right| ds' \int\limits_S \left|U_\beta^i(s) - G^i(s')\right| ds' \\ &+ \frac{2e^{8(2\gamma_2+m)^2R^2/\beta}}{\beta\pi^2} \int\limits_S \left|U_\beta^i(s) - G^i(s')\right| \sum_{k=1}^{m+2} \gamma_2^k \left|U_\beta^k(s) - G^k(s')\right| \left|DU_\beta^k(s)\right| ds' \end{aligned}$$

$$
\begin{aligned}
&+ \quad \frac{e^{8(2\gamma_2+m)^2R^2/\beta}}{\pi^2}\int\limits_S \left|DU^i_\beta(s)\right| ds' \\
&\leq \quad \frac{16R^2}{\beta\pi}e^{12(2\gamma_2+m)^2R^2/\beta}\sum_{k=1}^{m+2}\gamma_2^k\left|DU^k_\beta(s)\right| + \frac{8R^2}{\beta\pi}e^{8(2\gamma_2+m)^2R^2/\beta}\sum_{k=1}^{m+2}\gamma_2^k\left|DU^k_\beta(s)\right| \\
&+ \quad \frac{1}{\pi}e^{8(2\gamma_2+m)^2R^2/\beta}\left|DU^i_\beta(s)\right| = \sum_{k=1}^{m+2}C_k\left|DU^k_\beta(s)\right|.
\end{aligned}
$$

Finally, we conclude that

$$
\int\limits_S \left|Df^i(U_\beta)(s)\right|^2 ds \leq (m+2)\sum_{k=1}^{m+2}C_k^2\int\limits_S\left|DU^k_\beta(s)\right|^2 ds,
$$

and thus,

$$
f^i(U_\beta)\in \mathbb{H}^1(S), \quad i=1,\dots,m+2.
$$

From the Theorem 4.2 one can obtain that

$$
U^i_\beta(s)\in \mathbb{H}^3_{loc}(S), \quad i=1,\dots,m+2.
$$

Eventually, Theorem 4.3 implies $U^i_\beta\in C^1(S)$. Thus, $U^i_\beta\in \mathbb{H}^2_{loc}(S)\cap C^1(S)$.

Let show now that for all $s'\in S$ $e^{-|U_\beta(\cdot)-G(\cdot')|^2_{\gamma_2}/\beta}$ is in $C^1(S)$. Basing on the fact that $U^i_\beta\in C^1(S)$ we conclude that the first derivative

$$
D_s e^{-|U_\beta(s)-G(s')|^2_{\gamma_2}/\beta} = -\frac{2}{\beta}e^{-|U_\beta(s)-G(s')|^2_{\gamma_2}/\beta}\sum_{k=1}^{m+2}\gamma_2^k\left(U^k_\beta(s)-G^k(s')\right)DU^k_\beta(s)
$$

is bounded:

$$
\left|D_s e^{-|U_\beta(s)-G(s')|^2_{\gamma_2}/\beta}\right| \leq \frac{4R}{\beta}\sum_{k=1}^{m+2}\gamma_2^k\left|DU^k_\beta(s)\right| = M_1.
$$

Thus, $e^{-|U_\beta(\cdot)-G(\cdot')|^2_{\gamma_2}/\beta}\in C^1(S)$ for all $s'\in S$.

Let show now that for all $s'\in S$ $e^{-|U_\beta(\cdot)-G(\cdot')|^2_{\gamma_2}/\beta}$ is in $\mathbb{H}^2_{loc}(S)$:

$$
\begin{aligned}
D^2_s e^{-|U_\beta(s)-G(s')|^2_{\gamma_2}/\beta} &= -\frac{2}{\beta}D_s e^{-|U_\beta(s)-G(s')|^2_{\gamma_2}/\beta}\sum_{k=1}^{m+2}\gamma_2^k\left(U^k_\beta(s)-G^k(s')\right)DU^k_\beta(s) \\
&- \frac{2}{\beta}e^{-|U_\beta(s)-G(s')|^2_{\gamma_2}/\beta}\sum_{k=1}^{m+2}\gamma_2^k\left(DU^k_\beta(s)\right)^2 \\
&- \frac{2}{\beta}e^{-|U_\beta(s)-G(s')|^2_{\gamma_2}/\beta}\sum_{k=1}^{m+2}\gamma_2^k\left(U^k_\beta(s)-G^k(s')\right)D^2U^k_\beta(s).
\end{aligned}
$$

$$
\begin{aligned}
\left|D^2_s e^{-|U_\beta(s)-G(s')|^2_{\gamma_2}/\beta}\right| &\leq \frac{2}{\beta}\sum_{k=1}^{m+2}\gamma_2^k\left(2RM_1\left|DU^k_\beta(s)\right| + \left|DU^k_\beta(s)\right|^2\right) + \frac{4R}{\beta}\sum_{k=1}^{m+2}\gamma_2^k\left|D^2U^k_\beta(s)\right| \\
&= C_0 + \sum_{k=1}^{m+2}C_k\left|D^2U^k_\beta(s)\right|.
\end{aligned}
$$

Finally, we conclude that for all $V \subset\subset S$

$$\int\limits_V \left| D_s^2 e^{-|U_\beta(s)-G(s')|^2_{\gamma_2}/\beta} \right|^2 ds \leq (m+3) \left(C_0^2 \pi + \sum_{k=1}^{m+2} C_k^2 \int\limits_V \left| D^2 U_\beta^k(s) \right|^2 ds \right),$$

and thus,

$$e^{-|U_\beta(\cdot)-G(\cdot')|^2_{\gamma_2}/\beta} \in \mathbb{H}^2_{loc}(S), \quad \text{for all } s' \in S.$$

Let show now that $\Xi_{\beta,\gamma_2}(U_\beta)$ is in $C^1(S)$. Basing on the fact that for all $s' \in S$ $e^{-|U_\beta(\cdot)-G(\cdot')|^2_{\gamma_2}/\beta}$ is in $C^1(S)$, we conclude that the first derivative

$$D\Xi_{\beta,\gamma_2}(U_\beta)(s) = -\left(\Xi_{\beta,\gamma_2}(U_\beta)(s)\right)^2 \int\limits_S D_s e^{-|U_\beta(s)-G(s')|^2_{\gamma_2}/\beta} ds'$$

is bounded:

$$|D\Xi_{\beta,\gamma_2}(U_\beta)(s)| \leq \pi \left|\Xi_{\beta,\gamma_2}(U_\beta)(s)\right|^2 \left| D_s e^{-|U_\beta(s)-G(s')|^2_{\gamma_2}/\beta} \right| \leq \frac{1}{\pi} e^{8(2\gamma_2+m)^2R^2/\beta} \cdot M_1 = M_2.$$

Thus, $\Xi_{\beta,\gamma_2}(U_\beta) \in C^1(S)$.

Let show now that $\Xi_{\beta,\gamma_2}(U_\beta)$ is in $\mathbb{H}^2(S)$:

$$\begin{aligned} D^2\Xi_{\beta,\gamma_2}(U_\beta)(s) &= -2\Xi_{\beta,\gamma_2}(U_\beta)(s) \cdot D\Xi_{\beta,\gamma_2}(U_\beta)(s) \int\limits_S D_s e^{-|U_\beta(s)-G(s')|^2_{\gamma_2}/\beta} ds' \\ &- \left(\Xi_{\beta,\gamma_2}(U_\beta)(s)\right)^2 \int\limits_S D_s^2 e^{-|U_\beta(s)-G(s')|^2_{\gamma_2}/\beta} ds'. \end{aligned}$$

$$\begin{aligned} \left| D^2\Xi_{\beta,\gamma_2}(U_\beta)(s) \right| &\leq 2e^{4(2\gamma_2+m)^2R^2/\beta} M_1 M_2 + \frac{1}{\pi} e^{8(2\gamma_2+m)^2R^2/\beta} \left(C_0 + \sum_{k=1}^{m+2} C_k \left| D^2 U_\beta^k(s) \right| \right) \\ &= B_0 + \sum_{k=1}^{m+2} B_k \left| D^2 U_\beta^k(s) \right|. \end{aligned}$$

Finally, we conclude that for all $V \subset\subset S$

$$\int\limits_V \left| D^2\Xi_{\beta,\gamma_2}(U_\beta)(s) \right|^2 ds \leq (m+3) \left(B_0^2 \pi + \sum_{k=1}^{m+2} B_k^2 \int\limits_V \left| D^2 U_\beta^k(s) \right|^2 ds \right),$$

and thus,

$$\Xi_{\beta,\gamma_2}(U_\beta) \in \mathbb{H}^2_{loc}(S).$$

■

Appendix B

Proof of Proposition 4.3

We show at first that $f^i(U_\beta) \in \mathbb{H}^n_{loc}(S),\ i = 1, \ldots, m+2$. From the definition of $f^i(U_\beta)(s)$ we have

$$\begin{aligned}
D^n f^i(U_\beta)(s) &= \sum_{r_1+r_2+r_3+r_4=n} \frac{n!}{r_1!r_2!r_3!r_4!} D^{r_1}\Xi_{\beta,\gamma_2}(U_\beta)(s) D^{r_2}\Xi_{\beta,\gamma_2}(U_\beta)(s) \\
&\cdot \int_S D^{r_3} e^{-|U_\beta(s)-G(s')|^2_{\gamma_2}/\beta} D^{r_4}\left(U^i_\beta(s) - G^i(s')\right) ds' \\
&= \sum_{r_1+r_2+r_3+r_4=n,\ r_1,r_2,r_3,r_4\neq n} \frac{n!}{r_1!r_2!r_3!r_4!} D^{r_1}\Xi_{\beta,\gamma_2}(U_\beta)(s) D^{r_2}\Xi_{\beta,\gamma_2}(U_\beta)(s) \\
&\cdot \int_S D^{r_3} e^{-|U_\beta(s)-G(s')|^2_{\gamma_2}/\beta} D^{r_4}\left(U^i_\beta(s) - G^i(s')\right) ds' \\
&+ 2\Xi_{\beta,\gamma_2}(U_\beta)(s) D^n \Xi_{\beta,\gamma_2}(U_\beta)(s) \int_S e^{-|U_\beta(s)-G(s')|^2_{\gamma_2}/\beta}\left(U^i_\beta(s) - G^i(s')\right) ds' \\
&+ \left(\Xi_{\beta,\gamma_2}(U_\beta)(s)\right)^2 \int_S D^n_s e^{-|U_\beta(s)-G(s')|^2_{\gamma_2}/\beta}\left(U^i_\beta(s) - G^i(s')\right) ds' \\
&+ \left(\Xi_{\beta,\gamma_2}(U_\beta)(s)\right)^2 D^n U^i_\beta(s) \int_S e^{-|U_\beta(s)-G(s')|^2_{\gamma_2}/\beta} ds'
\end{aligned}$$

Using the estimate (4.11) and taking into account that $e^{-|U(s)-G(s')|^2_{\gamma_2}/\beta} \leq 1$ and $|U^i_\beta(s) - G^i(s')| \leq 2R$, we obtain

$$\left|D^n f^i(U_\beta)(s)\right| \leq C_0 + C_1 \left|D^n \Xi_{\beta,\gamma_2}(U_\beta)(s)\right| + C_2 \left|D^n_s e^{-|U_\beta(s)-G(s')|^2_{\gamma_2}/\beta}\right| + C_3 \left|D^n U^i_\beta(s)\right|.$$

Finally, we conclude that for every $V \subset\subset S$

$$\begin{aligned}
\int_V \left|D^n f^i(U_\beta)(s)\right|^2 ds &\leq 4C_0^2\pi + 4C_1^2 \int_V \left|D^n \Xi_{\beta,\gamma_2}(U_\beta)(s)\right|^2 ds \\
&+ 4C_2^2 \int_V \left|D^n_s e^{-|U_\beta(s)-G(s')|^2_{\gamma_2}/\beta}\right|^2 ds + 4C_3^2 \int_V \left|D^n U^i_\beta(s)\right|^2 ds,
\end{aligned}$$

and thus, $f^i(U_\beta) \in \mathbb{H}^n_{loc}(S)$ for $i = 1, \dots, m+2$.

From the Theorem 4.2 one can obtain that

$$U^i_\beta \in \mathbb{H}^{n+2}_{loc}(S), \quad i = 1, \dots, m+2.$$

Theorem 4.3 implies $U^i_\beta \in C^n(S)$. Thus, $U^i_\beta \in \mathbb{H}^{n+1}_{loc}(S) \cap C^n(S)$.

Let show now that for all $s' \in S$ $e^{-|U_\beta(\cdot)-G(\cdot')|^2_{\gamma_2}/\beta}$ is in $C^n(S)$. Basing on the fact that $U^i_\beta \in C^n(S)$ we conclude that the n-th derivative

$$\begin{aligned} D^n_s e^{-|U_\beta(s)-G(s')|^2_{\gamma_2}/\beta} &= D^{n-1}_s \left(-\frac{2}{\beta} e^{-|U_\beta(s)-G(s')|^2_{\gamma_2}/\beta} \sum_{k=1}^{m+2} \gamma_2^k \left(U^k_\beta(s) - G^k(s') \right) DU^k_\beta(s) \right) \\ &= -\frac{2}{\beta} \sum_{r_1+r_2+r_3=n-1} \frac{(n-1)!}{r_1!r_2!r_3!} D^{r_1}_s e^{-|U_\beta(s)-G(s')|^2_{\gamma_2}/\beta} \\ &\cdot \sum_{k=1}^{m+2} \gamma_2^k D^{r_2} \left(U^k_\beta(s) - G^k(s') \right) D^{r_3+1} U^k_\beta(s) \end{aligned}$$

is bounded:

$$\begin{aligned} \left| D^n_s e^{-|U_\beta(s)-G(s')|^2_{\gamma_2}/\beta} \right| &\le \frac{2}{\beta} \sum_{r_1+r_2+r_3=n-1} \frac{(n-1)!}{r_1!r_2!r_3!} \left| D^{r_1}_s e^{-|U_\beta(s)-G(s')|^2_{\gamma_2}/\beta} \right| \\ &\cdot \sum_{k=1}^{m+2} \gamma_2^k \left| D^{r_2} \left(U^k_\beta(s) - G^k(s') \right) \right| \left| D^{r_3+1} U^k_\beta(s) \right| = M_3. \end{aligned}$$

Thus, $e^{-|U_\beta(\cdot)-G(\cdot')|^2_{\gamma_2}/\beta} \in C^n(S)$ for all $s' \in S$.

Let show now that for all $s' \in S$ $e^{-|U_\beta(\cdot)-G(\cdot')|^2_{\gamma_2}/\beta}$ is in $\mathbb{H}^{n+1}_{loc}(S)$:

$$\begin{aligned} D^{n+1}_s e^{-|U_\beta(s)-G(s')|^2_{\gamma_2}/\beta} &= -\frac{2}{\beta} \sum_{r_1+r_2+r_3=n,\ r_3 \neq n} \frac{n!}{r_1!r_2!r_3!} D^{r_1}_s e^{-|U_\beta(s)-G(s')|^2_{\gamma_2}/\beta} \\ &\cdot \sum_{k=1}^{m+2} \gamma_2^k D^{r_2} \left(U^k_\beta(s) - G^k(s') \right) D^{r_3+1} U^k_\beta(s) \\ &- \frac{2}{\beta} e^{-|U_\beta(s)-G(s')|^2_{\gamma_2}/\beta} \cdot \sum_{k=1}^{m+2} \gamma_2^k \left(U^k_\beta(s) - G^k(s') \right) D^{n+1} U^k_\beta(s). \end{aligned}$$

$$\begin{aligned} \left| D^{n+1}_s e^{-|U_\beta(s)-G(s')|^2_{\gamma_2}/\beta} \right| &\le \frac{2}{\beta} \sum_{r_1+r_2+r_3=n,\ r_3 \neq n} \frac{n!}{r_1!r_2!r_3!} \left| D^{r_1}_s e^{-|U_\beta(s)-G(s')|^2_{\gamma_2}/\beta} \right| \\ &\cdot \sum_{k=1}^{m+2} \gamma_2^k \left| D^{r_2} \left(U^k_\beta(s) - G^k(s') \right) \right| \cdot \left| D^{r_3+1} U^k_\beta(s) \right| \\ &+ \frac{4R}{\beta} \sum_{k=1}^{m+2} \gamma_2^k \left| D^{n+1} U^k_\beta(s) \right| = C_0 + \sum_{k=1}^{m+2} C_k \left| D^{n+1} U^k_\beta(s) \right|. \end{aligned}$$

Finally, we conclude that for all $V \subset\subset S$

$$\int\limits_V \left| D_s^{n+1} e^{-|U_\beta(s)-G(s')|^2_{\gamma_2}/\beta} \right|^2 ds \leq (m+3) \left(C_0^2 \pi + \sum_{k=1}^{m+2} C_k^2 \int\limits_V \left| D^{n+1} U_\beta^k(s) \right|^2 ds \right),$$

and thus, $e^{-|U_\beta(\cdot)-G(\cdot')|^2_{\gamma_2}/\beta} \in \mathbb{H}^{n+1}(S)$ for all $s' \in S$.

Let show now that $\Xi_{\beta,\gamma_2}(U_\beta)$ is in $C^n(S)$. Basing on the fact that for all $s' \in S$ $e^{-|U_\beta(\cdot)-G(\cdot')|^2_{\gamma_2}/\beta}$ is in $C^n(S)$, we conclude that the n-th derivative

$$\begin{aligned} D^n \Xi_{\beta,\gamma_2}(U_\beta)(s) &= D^{n-1} \left(-\left(\Xi_{\beta,\gamma_2}(U_\beta)(s)\right)^2 \int\limits_S D_s e^{-|U_\beta(s)-G(s')|^2_{\gamma_2}/\beta} ds' \right) \\ &= -\sum_{r_1+r_2+r_3=n-1} \frac{(n-1)!}{r_1! r_2! r_3!} D^{r_1} \Xi_{\beta,\gamma_2}(U_\beta)(s) D^{r_2} \Xi_{\beta,\gamma_2}(U_\beta)(s) \\ &\cdot \int\limits_S D_s^{r_3+1} e^{-|U_\beta(s)-G(s')|^2_{\gamma_2}/\beta} ds' \end{aligned}$$

is bounded:

$$\begin{aligned} |D^n \Xi_{\beta,\gamma_2}(U_\beta)(s)| &\leq \pi \sum_{r_1+r_2+r_3=n-1} \frac{(n-1)!}{r_1! r_2! r_3!} |D^{r_1} \Xi_{\beta,\gamma_2}(U_\beta)(s)| \cdot |D^{r_2} \Xi_{\beta,\gamma_2}(U_\beta)(s)| \\ &\cdot \left| D_s^{r_3+1} e^{-|U_\beta(s)-G(s')|^2_{\gamma_2}/\beta} \right|. \end{aligned}$$

Thus, $\Xi_{\beta,\gamma_2}(U_\beta) \in C^n(S)$.

Let show now that $\Xi_{\beta,\gamma_2}(U_\beta)$ is in $\mathbb{H}^{n+1}_{loc}(S)$:

$$\begin{aligned} D^{n+1} \Xi_{\beta,\gamma_2}(U_\beta)(s) &= -\sum_{r_1+r_2+r_3=n,\ r_3 \neq n} \frac{n!}{r_1! r_2! r_3!} D^{r_1} \Xi_{\beta,\gamma_2}(U_\beta)(s) \cdot D^{r_2} \Xi_{\beta,\gamma_2}(U_\beta)(s) \\ &\cdot \int\limits_S D_s^{r_3+1} e^{-|U_\beta(s)-G(s')|^2_{\gamma_2}/\beta} ds' - \left(\Xi_{\beta,\gamma_2}(U_\beta)(s)\right)^2 \\ &\cdot \int\limits_S D_s^{n+1} e^{-|U_\beta(s)-G(s')|^2_{\gamma_2}/\beta} ds'. \end{aligned}$$

$$\begin{aligned} \left| D^{n+1} \Xi_{\beta,\gamma_2}(U_\beta)(s) \right| &\leq \pi \sum_{r_1+r_2+r_3=n,\ r_3 \neq n} \frac{n!}{r_1! r_2! r_3!} |D^{r_1} \Xi_{\beta,\gamma_2}(U_\beta)(s)| \cdot |D^{r_2} \Xi_{\beta,\gamma_2}(U_\beta)(s)| \\ &\cdot \left| D_s^{r_3+1} e^{-|U_\beta(s)-G(s')|^2_{\gamma_2}/\beta} \right| + \frac{1}{\pi} e^{8(2\gamma_2+m)^2 R^2/\beta} \cdot \left| D_s^{n+1} e^{-|U_\beta(s)-G(s')|^2_{\gamma_2}/\beta} \right| \\ &= C_0 + C_1 \left| D_s^{n+1} e^{-|U_\beta(s)-G(s')|^2_{\gamma_2}/\beta} \right|. \end{aligned}$$

Finally, we conclude that for every $V \subset\subset S$

$$\int\limits_V \left| D^{n+1} \Xi_{\beta,\gamma_2}(U_\beta)(s) \right|^2 ds \leq 2 \left(C_0^2 \pi + C_1^2 \int\limits_V \left| D_s^{n+1} e^{-|U_\beta(s)-G(s')|^2_{\gamma_2}/\beta} \right|^2 ds \right),$$

and thus, $\Xi_{\beta,\gamma_2}(U_\beta)(s) \in \mathbb{H}^{n+1}_{loc}(S)$. ■

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